Supplementary Information

GraphPathology enables geometry-aware reconstruction of spatial transcriptomics from sparse molecular measurements

Truong-Son Hy[1,*]

[1]Department of Computer Science, University of Alabama at Birmingham, Birmingham, Alabama, United States

[*]Correspondence: thy@uab.edu

Contents and source-data guide

This document provides supporting figures and tables, cohort-resolved graph comparisons, computational diagnostics and detailed development methods. Source tables retain all tested estimators, masks, training seeds and defined-correlation counts. Patients are biological replicates; masks and training seeds describe computational variation. The main Methods distinguish the original evaluation, retrospective controls and the two additional neural training regimes.

S1 Supplementary notes

S1.1 Affine interpolation identity

For a gene with transform $T(y) = (y - \mu)/\sigma$, $\sigma > 0$, and interpolation weights satisfying $\sum_j w_{ij} = 1$, we have

$$T^{-1}\left(\sum_j w_{ij} T(y_j)\right) = \sigma \sum_j w_{ij} \frac{y_j - \mu}{\sigma} + \mu$$
$$= \sum_j w_{ij} y_j.$$

Consequently, applying and inverting a fixed per-gene affine transform does not create a distinct molecular interpolation method when weights are unchanged. In the Wu evaluation, the raw and transformed predictions differed by at most 2.66×10^{-15}, consistent with roundoff. Tiny numerical changes can break tied ranks and alter raw Spearman calculations; such changes were logged as numerical rank sensitivity, not counted as harmonization benefits. Calibration can still alter an image branch or a learned fusion model; that is a different comparison. The manuscript therefore omits duplicate molecular interpolation curves and makes no claim of an affine-harmonization advantage for molecular interpolation itself.

S1.2 Dataset roles, masking and interpretation

The four-patient HEST cohort was used for development. HER2ST was first evaluated under a frozen protocol and subsequently used for calibration development; those two views reuse the same eight patients. The Wu cohort supplied three calibration and three evaluation patients. The 22 Bassiouni patients were evaluated without new cohort-specific fitting. Counts of independent evaluation patients must not add repeated HER2ST analyses or mask replicates as new patients.

Distributed evaluation reveals molecular vectors at sampled anchors and scores all remaining locations. The external regional-completion protocol instead withholds a contiguous 40% region, samples anchors outside it, and scores that region only. V1 uses inverse-distance-squared interpolation and V2 uses an equal-weight nearest-anchor mean. Their contrast is therefore a protocol-level difference, not an isolated geometry intervention. The earlier development block/capture masks are also not identical to the external forbidden-region design.

Public external H&E images are downsampled, and sparse molecular interpolation receives same-section expression unavailable to the image-only baseline. Image-resolution limitations and unequal available information must accompany comparisons with histology. An H&E-fusion failure under these frozen inputs does not establish that native-resolution histology is generally uninformative.

S1.3 Aggregation and metric interpretation

The primary external summaries compute gene-level metrics on hidden locations, summarize genes within each run, average random seeds within geometry, average geometries within pa-

tient, and finally summarize patients. Unless a figure legend states otherwise, external cohort markers are medians across patients. Several development and transportability plots instead show arithmetic means; their legends state this explicitly. A difference between two cohort medians is not a median paired patient effect.

Mean squared error and mean absolute error use the unchanged development-standardized log1p-count target. Shared numerical units do not remove capture-depth or assay differences. Correlations are undefined when the target or prediction range is at most 10^{-12}; they are excluded from correlation averages but not counted as successes in threshold fractions. Spearman correlations use average ranks for ties. Mask and model seeds describe computational variation, not independent biological replication. Descriptive patient-bootstrap intervals are reported for the 22-patient comparisons (main Tables 3–5); they are not p-values. Wu patients are reported individually, with no primary three-patient bootstrap claim.

S2 Supplementary figures

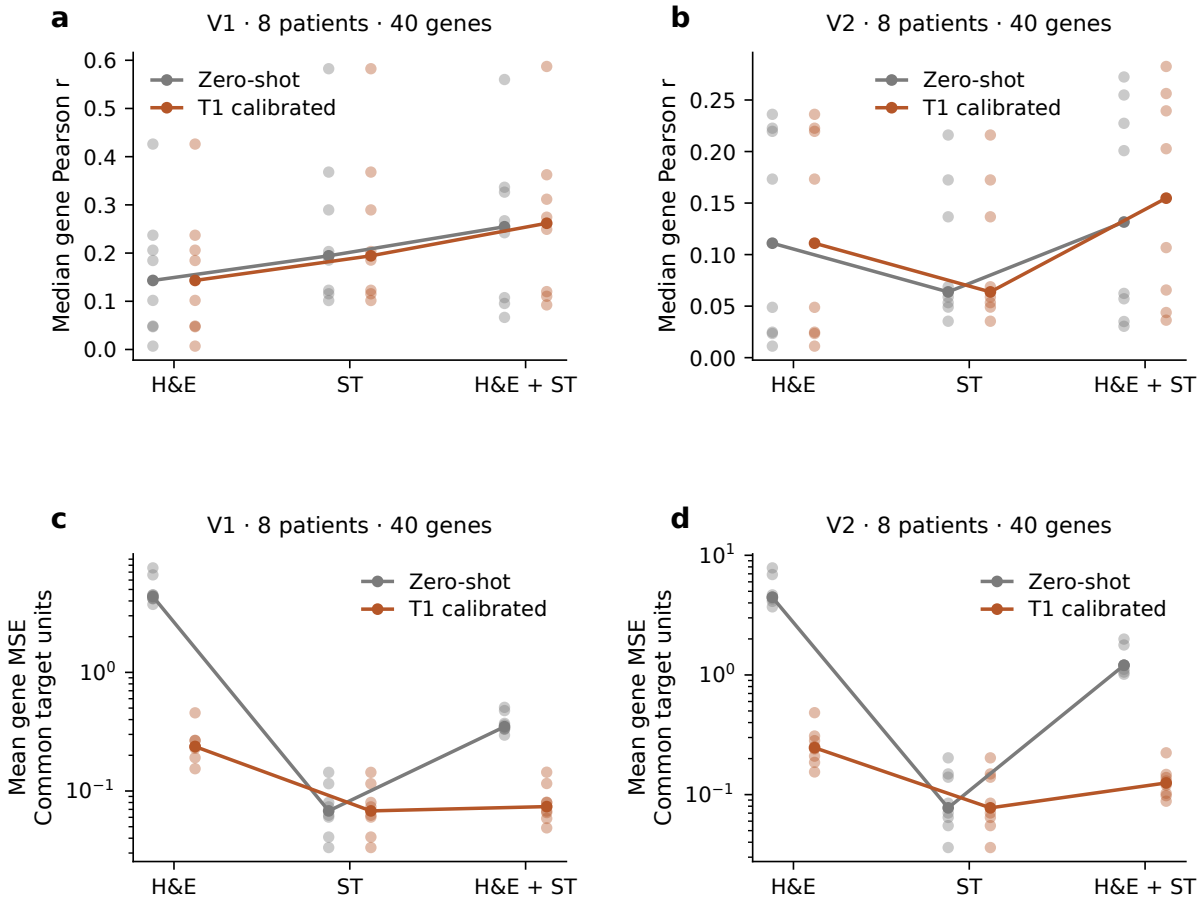


Figure S1: HER2ST transportability and target calibration. Eight HER2ST patients and the frozen 40-gene intersection are shown at 20% random measured-location coverage. **a,c,** Distributed interpolation (V1). **b,d,** Contiguous regional completion (V2). Panels a/b show median gene Pearson correlation and c/d show mean gene MSE in common target units on a logarithmic axis. Dots are patient summaries after averaging seeds and geometries; connected markers are medians across patients. T0 denotes the uncalibrated zero-shot analysis, whereas T1 denotes the calibrated target mapping with fixed fusion. Image weights were 0.25 for V1 and 0.5 for V2; these development weights differ from the final external weight of 0.2. These are the same patients, not independent replication groups. The T1 fixed-fusion analysis differs from the later fold-selected fusion results in Supplementary Fig. S10. Affine transformation and inverse transformation leave molecular interpolation unchanged; apparent image/fusion calibration effects are not a new interpolation advantage.

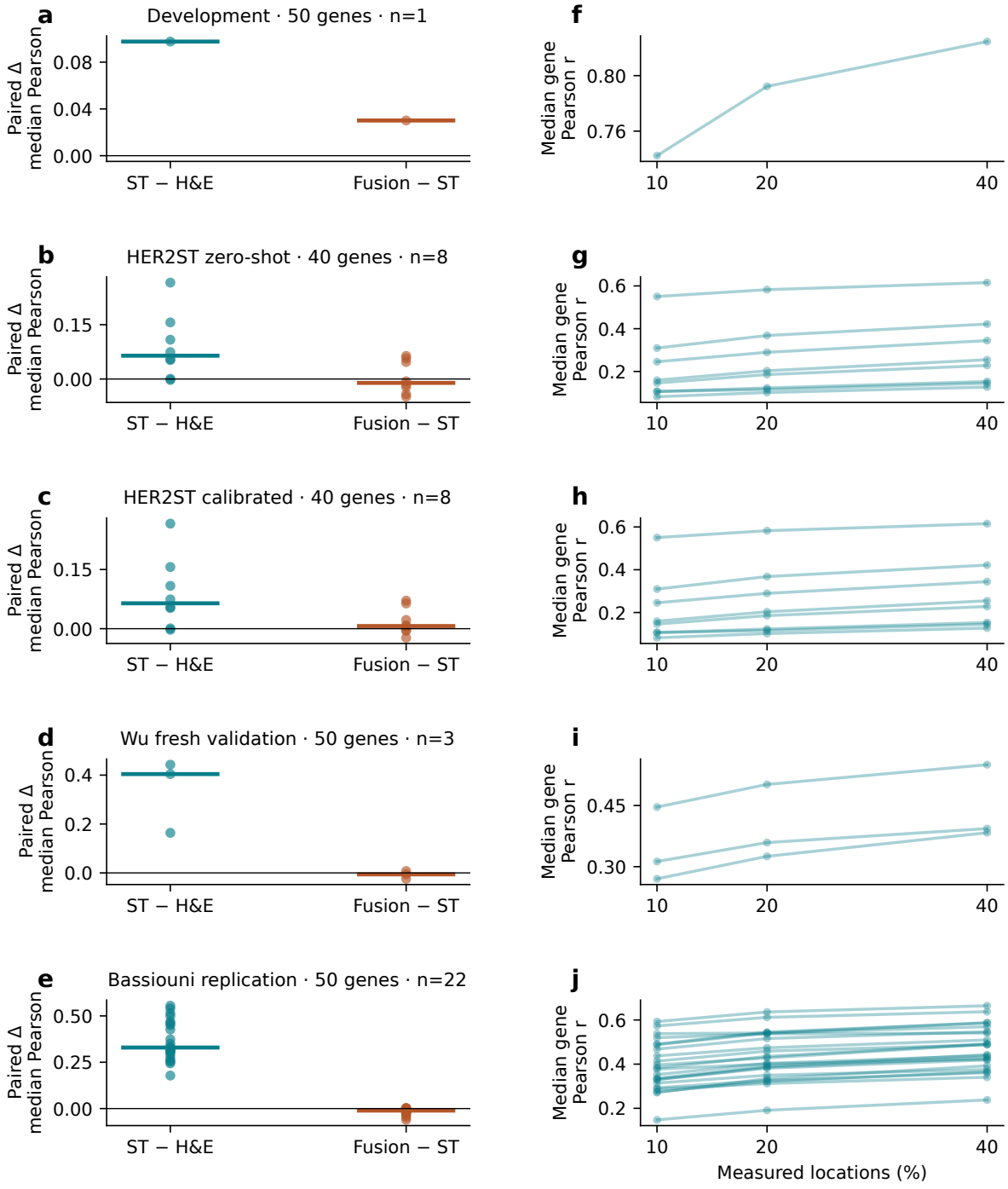


Figure S2: Cross-cohort synthesis without pooling different panels. Rows correspond to the development primary patient ($n = 1$), HER2ST zero-shot ($n = 8$), HER2ST calibrated ($n = 8$, the same patients), original Wu evaluation ($n = 3$) and original Bassiouni evaluation ($n = 22$). The development and original evaluation cohorts use 50 genes; HER2ST uses 40. **a–e,** Left-column paired patient correlation effects at 20% distributed coverage for molecular interpolation minus image-only prediction and fusion minus molecular interpolation; bars are median paired effects. **f–j,** Right-column individual patient molecular correlation curves at 10%, 20% and 40% distributed coverage. The panels preserve differences in analytical role and gene panel and do not define a pooled cross-cohort score. HER2ST calibration is development, not an additional independent validation cohort.

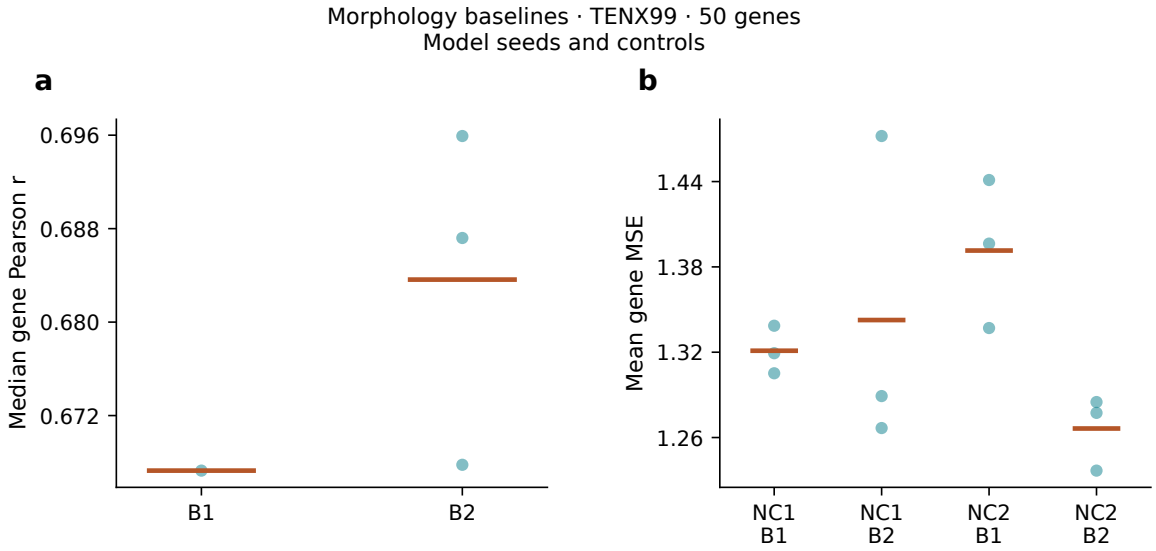


Figure S3: Morphology-only development baselines and controls. The morphology-only development analyses use the single primary development patient TENX99 and 50 genes. **a,** Median gene Pearson correlation for ridge regression (B1) and the small multilayer perceptron (B2). The training-mean baseline has constant predictions and undefined correlations and is not drawn as zero. **b,** MSE under training-target-row permutation (NC1) and training-embedding-row permutation (NC2). Dots represent saved model-seed runs and bars are arithmetic means of displayed runs, not independent patient estimates. The controls disrupt image–expression pairing while preserving validation and evaluation data. This initial image-only head differs from the frozen local MLP selected in the representation comparison and used for external replication.

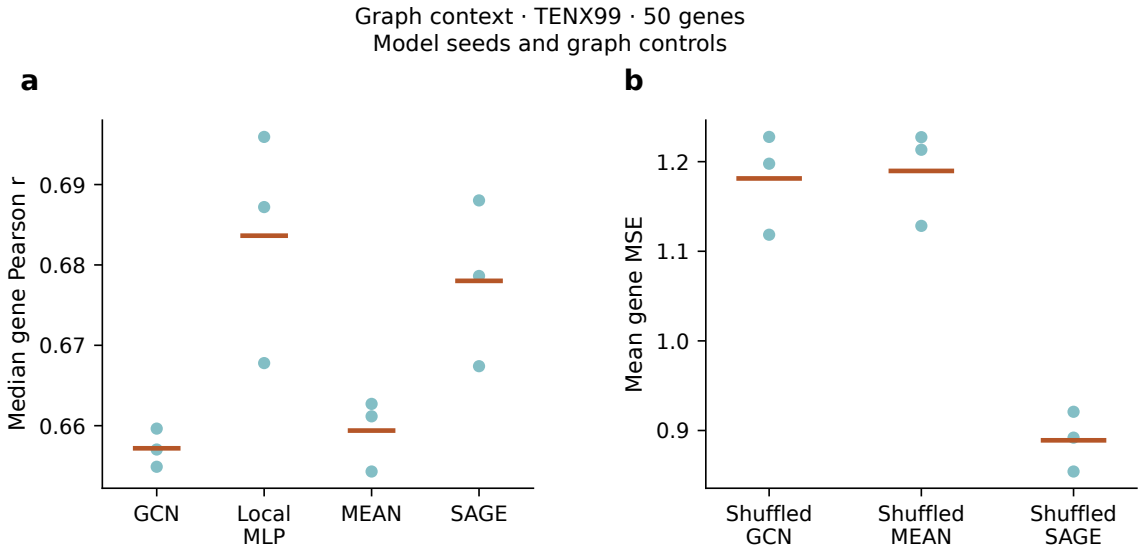


Figure S4: Single-scale graph context did not by itself establish spatial specificity. The single-scale graph comparison uses TENX99 and the 50-gene panel. **a,** Median gene correlation for the frozen local MLP, neighborhood mean pooling plus MLP, GCN and GraphSAGE. **b,** Mean gene MSE for the archived shuffled-neighborhood runs. Dots are training-seed runs and bars are their arithmetic means. A graph model outperforming a local predictor does not establish that the correct spatial adjacency caused the gain; generic aggregation or model parameterization can also contribute. The results use the original fitted models without additional graph selection or local-predictor refitting.

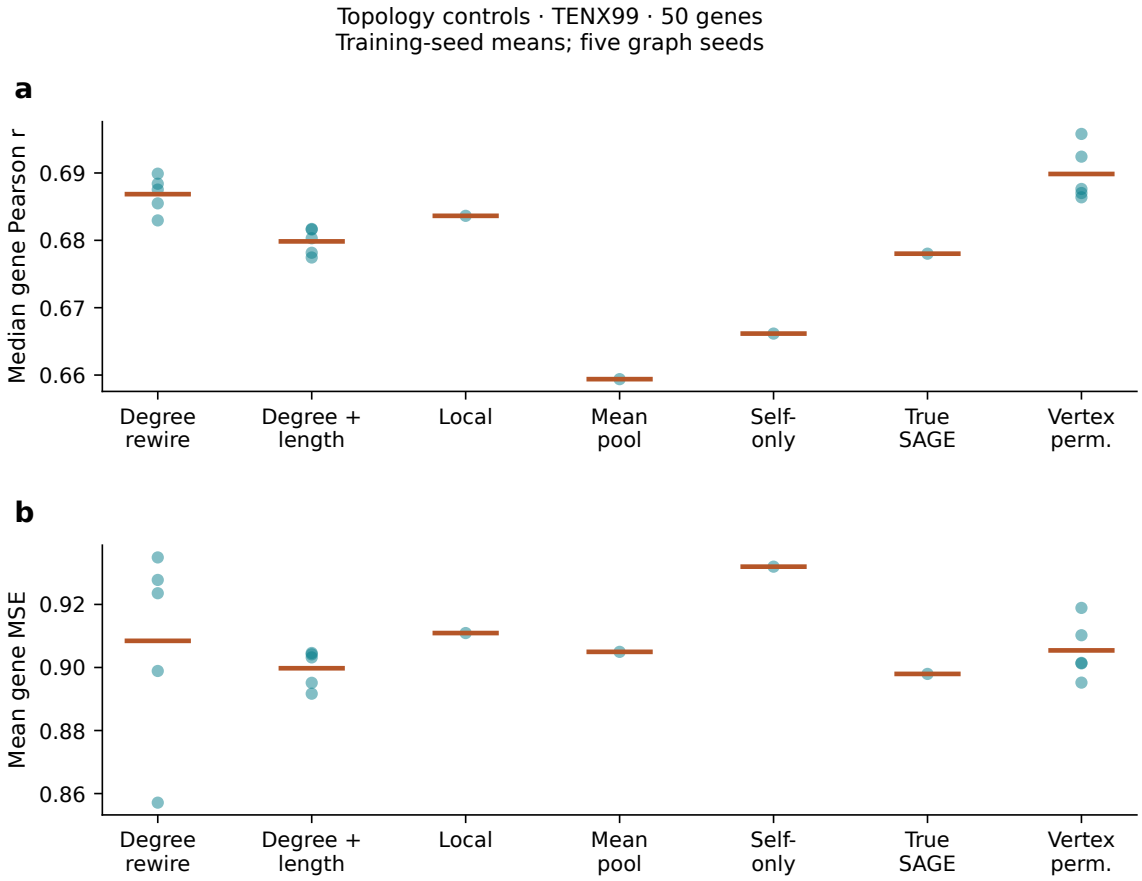


Figure S5: Matched graph controls limited claims about exact topology. The matched graph controls use one development patient, TENX99, and 50 genes. **a,** Median gene Pearson correlation. **b,** Mean gene MSE. Local, self-only, mean-pooling and true-spatial GraphSAGE results accompany vertex-permuted, degree-rewired and degree-plus-edge-length-rewired controls. Training seeds are averaged within each graph randomization seed; the randomized controls have five graph seeds. Dots are these averages and bars are arithmetic means. Random graph seeds are not biological replicates. The experiment did not establish a reproducible advantage of the true spatial topology. This figure records a negative/mixed development result, not evidence that topology is irrelevant in every representation or tissue.

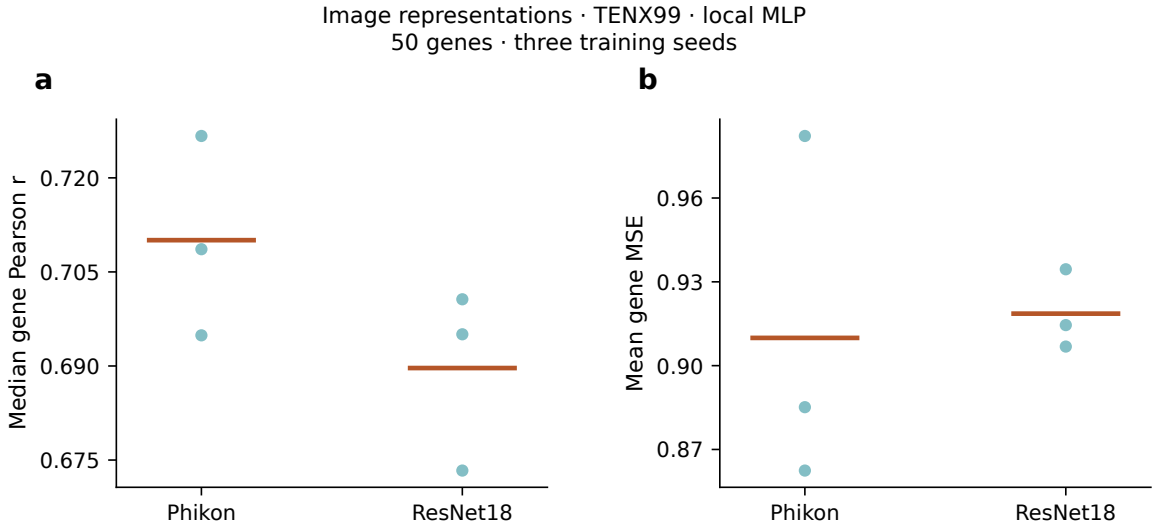


Figure S6: Pathology representation effects in the development patient. The local MLP representation comparison evaluates frozen ImageNet ResNet18 and Phikon representations in TENX99 for 50 genes. **a,** Median gene Pearson correlation. **b,** Mean gene MSE. Each dot is one of three training seeds; bars are arithmetic means. The panel-specific improvement did not establish general superiority on larger target panels or reproducible topology-specific gains. The original frozen Bassiouni evaluation retained the ResNet18 image branch. Phikon was subsequently evaluated in Wu and Bassiouni patients in the separate exploratory matched-backbone study. This plot remains a one-patient development comparison, not external encoder validation.

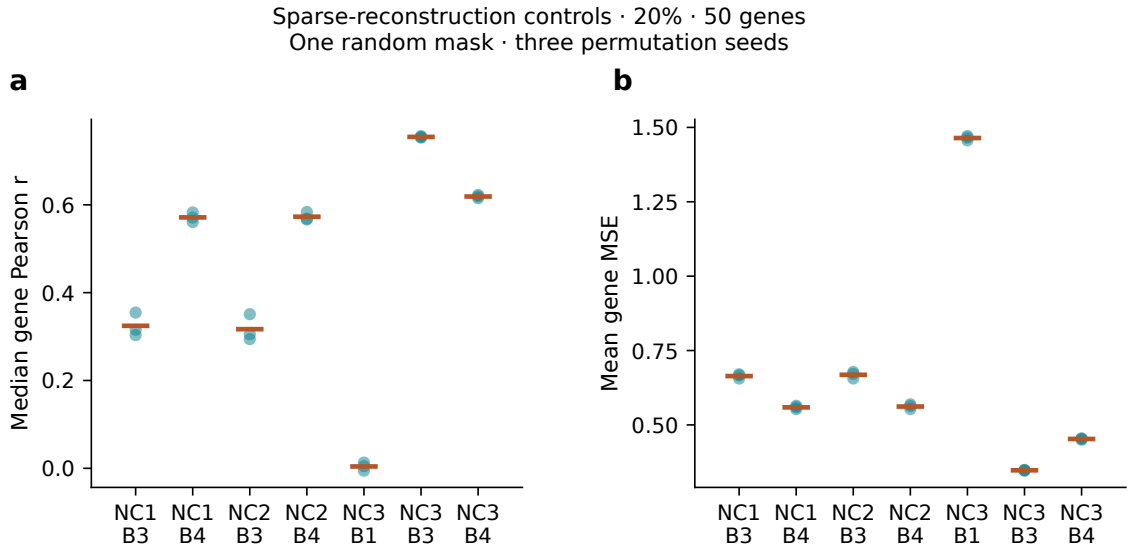


Figure S7: Sparse-reconstruction development controls. The sparse-reconstruction controls use the ResNet18 branch, TENX99, 50 genes and 20% measured-location coverage. In this figure, B1 denotes image-only prediction, B3 image–molecular fusion and B4 learned propagation. Controls include anchor-expression permutation (NC1), anchor-position permutation (NC2) and image-location permutation (NC3). Model labels are specific to this experiment. **a,** Median gene Pearson correlation. **b,** Mean gene MSE. All controls use the same random observation mask (replicate 0); dots represent three permutation seeds and bars are arithmetic means. These distributions are control diagnostics rather than estimates of a geometry effect. Separate integrity checks establish invariance to corruption of hidden labels. Related anchor-assignment disruptions must not be counted as independent biological mechanisms.

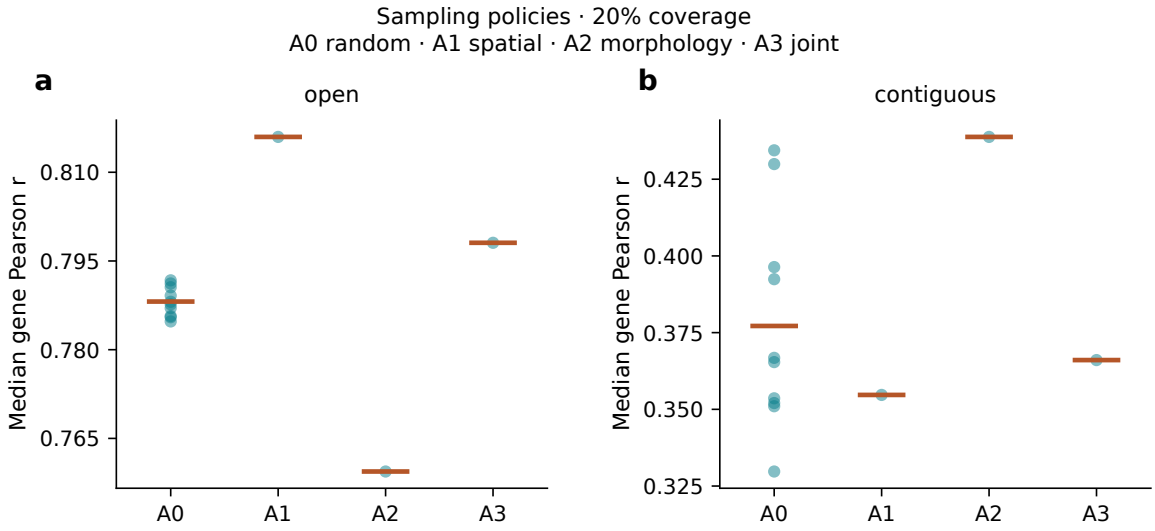


Figure S8: Sampling effects depended on the reconstruction regime. The sampling comparison uses interpolation-only results for TENX99, the ResNet18 reference branch, 50 genes and 20% coverage. **a,** Open-tissue interpolation. **b,** Contiguous missing-region evaluation. A0 is random sampling with ten seeds; A1 is spatial farthest-point sampling, A2 morphology-space farthest-point sampling, and A3 a locked joint policy. A1–A3 are deterministic given their stored initialization. Dots are policy runs and bars are arithmetic means. This limited baseline diagnostic is not a complete summary of all learned/fusion families and does not demonstrate universally optimal sampling.

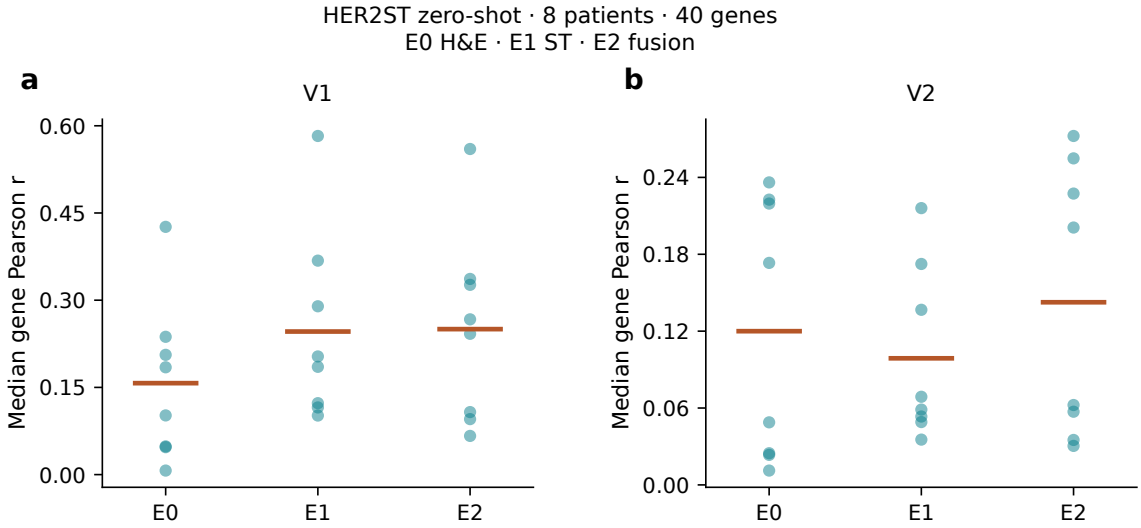


Figure S9: HER2ST patient distributions under the frozen zero-shot protocol. The eight HER2ST patients, 40 shared primary genes and 20% random-anchor coverage are summarized separately for **a,** distributed interpolation and **b,** contiguous regional completion. E0 denotes H&E-only prediction, E1 molecular interpolation and E2 frozen fusion, with image weights 0.25 for V1 and 0.5 for V2. Dots are patient means across seeds/geometries. Bars are arithmetic patient means, unlike the median summaries in Supplementary Fig. S1 and the primary tables. These zero-shot results and the subsequent calibration analyses use the same patients; HER2ST therefore does not supply an untouched final validation cohort.

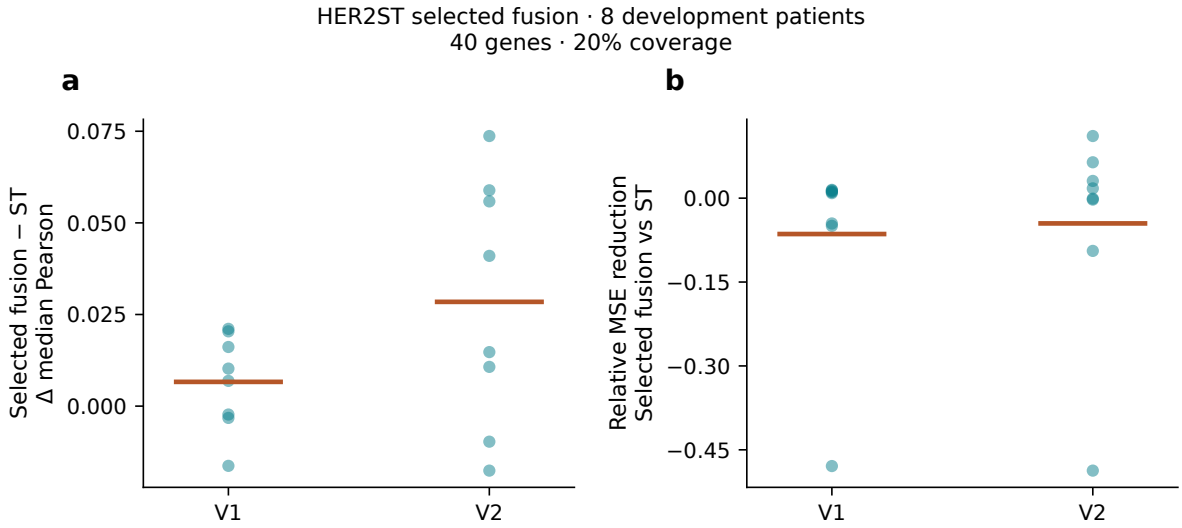


Figure S10: Patient-disjoint calibration did not establish robust fusion superiority. The calibration-development analysis includes eight HER2ST patients, the frozen 40-gene intersection and 20% coverage. **a,** Selected calibrated fusion minus molecular interpolation in Pearson correlation. **b,** Relative MSE reduction compared with molecular interpolation; positive values indicate lower fusion error. Dots are patient/fold results grouped by V1/V2 and bars are arithmetic means. The selected fusion rule was determined within patient-disjoint training/calibration folds, not from the displayed outer-patient outcomes. Joint correlation and error improvements were not consistent across patients. These selected-fusion results should not be interchanged with the T1 fixed-fusion target study shown in Supplementary Fig. S1.

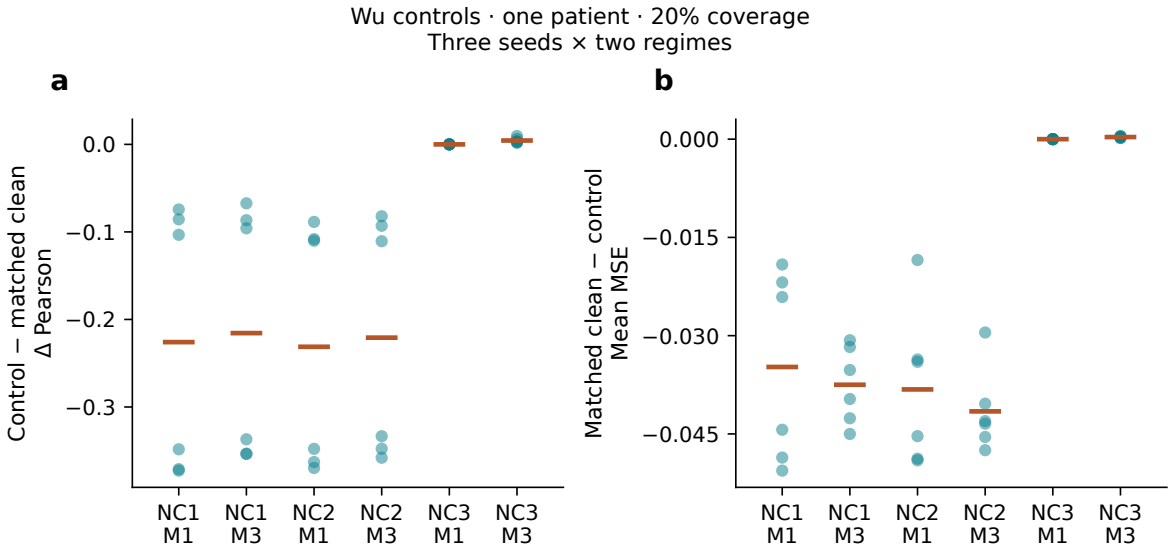


Figure S11: Matched negative controls in original Wu evaluation. The predeclared held-out patient 1160920F, 50 genes, 20% coverage and random mask seed 0 were used, with V1 geometry 0 and V2 geometry 1. Three permutation seeds were evaluated for NC1 anchor-vector reassignment, NC2 anchor-position reassignment and NC3 image-prediction permutation. **a,** Control minus matched-clean median gene Pearson correlation. **b,** Matched-clean minus control MSE, so negative values indicate increased control error. Molecular interpolation (M1) and fixed image fusion (M3) are shown. Dots are matched control settings and bars are means; they are not independent patients. Molecular controls disrupt the correct spatial anchor assignments, whereas image permutation provides no robust supporting evidence for the frozen image contribution.

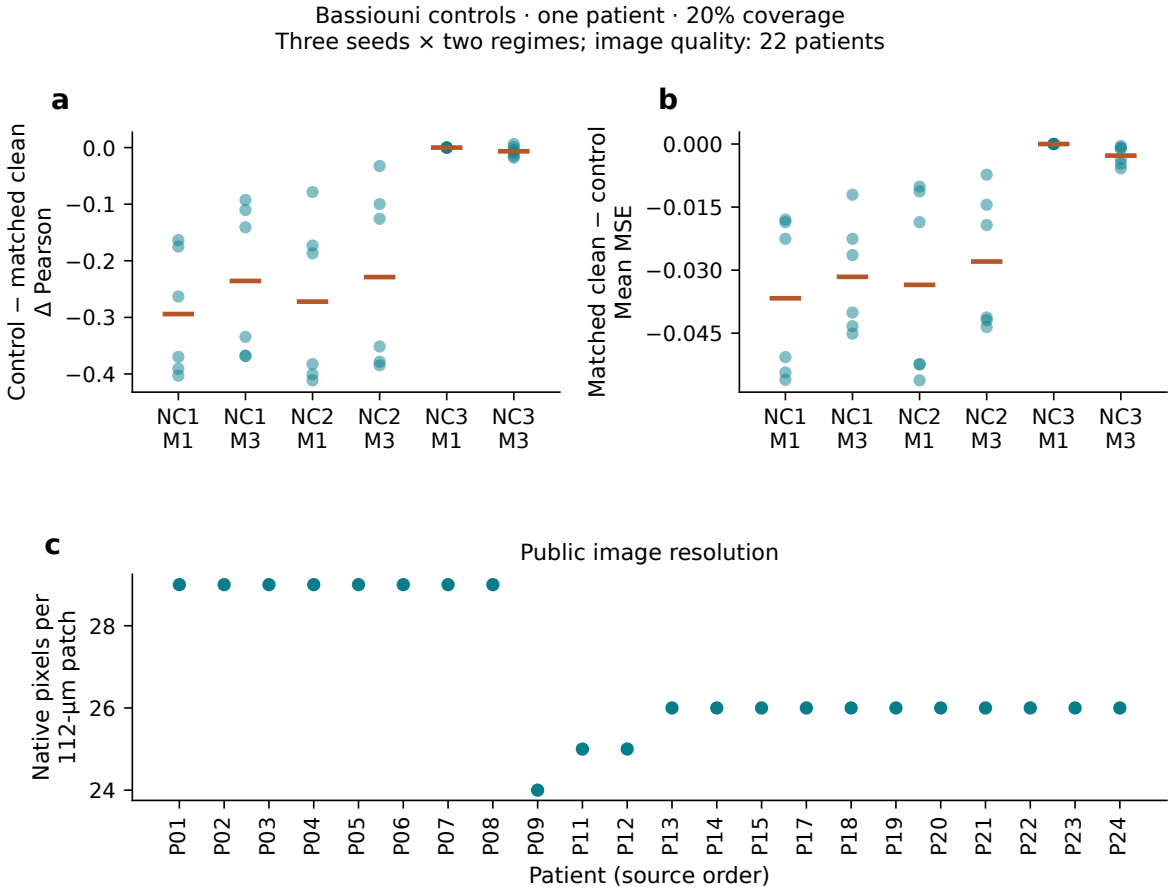


Figure S12: Replication controls and native image resolution. The independent-replication controls use the predeclared Bassiouni patient BASS_P01, 50 genes, 20% coverage, random mask seed 0, V1 geometry 0 and V2 geometry 1. **a,b,** Matched Pearson and MSE effects for M1/M3 under three control seeds, with definitions and signs as in Supplementary Fig. S11. Dots are matched control settings and bars are arithmetic means. **c,** Native pixels per nominal 112-μm patch for all 22 patients in source order. The supplied images provide only 24–29 native pixels before resizing for the encoder. This is image-input quality control, not expression-prediction performance. It explains why the image comparisons cannot establish the limits of native-resolution histopathology. Coordinate overlays for all patients are recorded in the archived qualification audit.

S3 Supplementary tables

The tables below provide compact views of the accompanying machine-readable data and definitions needed to interpret the supplementary figures. Table S2 is rounded for presentation; the digital files retain unrounded exported values. Large gene-level and distance-resolved tables are supplied as separate files rather than reproduced in full in this PDF. Mask, model and permutation seeds are computational replicates and do not increase the number of independent patients.

Table S1: Primary gene panel. The 50 genes are listed in the order fixed by variance ranking in the development-training patients. The original Wu and Bassiouni evaluation retain all 50 genes. HER2ST analyses use the 40-gene intersection with this panel; unavailable genes are not replaced. The full data file also includes the larger development panels and cohort-specific availability records: `source_data/SuppTable1_gene_panels.csv`.

Order	Gene	Order	Gene
1	FOXA1	26	LUM
2	SCD	27	MYO5B
3	KRT7	28	SFRP4
4	ERBB2	29	KRT14
5	POSTN	30	TOP2A
6	CEACAM6	31	IL7R
7	SERHL2	32	AR
8	ANKRD30A	33	KRT5
9	EPCAM	34	PCLAF
10	ABCC11	35	KLF5
11	TACSTD2	36	SFRP1
12	GATA3	37	TRAF4
13	FASN	38	CXCR4
14	MLPH	39	MYH11
15	S100A14	40	TENT5C
16	CDH1	41	TCIM
17	CD9	42	NARS
18	KRT8	43	ACTG2
19	CCND1	44	PDK4
20	DSP	45	CD3E
21	SERPINA3	46	TRAC
22	LYPD3	47	JUP
23	TFAP2A	48	LYZ
24	TPD52	49	CENPF
25	PTGDS	50	CCDC80

Table S2: Patient-resolved reconstruction metrics. Compact view of the 25 original external-evaluation patients at 20% distributed coverage, using 50 genes. Values average mask seeds within patient. r is the median gene Pearson correlation; MSE is mean gene squared error in the common target units. I denotes image-only, ST molecular interpolation, and F fixed image fusion. Complete exported patient summaries for the available cohorts, regimes, budgets and methods are supplied in `source_data/SuppTable2_per_patient.csv`.

Patient	r_{I}	r_{ST}	r_{F}	$\mathrm{MSE}_{\mathrm{I}}$	$\mathrm{MSE}_{\mathrm{ST}}$	$\mathrm{MSE}_{\mathrm{F}}$
Wu						
1160920F	-0.045	0.359	0.335	0.846	0.082	0.095
CID4290	0.161	0.325	0.333	2.915	0.099	0.100
CID44971	0.059	0.502	0.496	2.949	0.115	0.118
Bassiouni						
BASS_P01	0.077	0.382	0.370	1.304	0.087	0.106
BASS_P02	0.009	0.459	0.431	1.067	0.093	0.127
BASS_P03	0.030	0.545	0.537	2.585	0.081	0.089
BASS_P04	-0.039	0.334	0.312	1.882	0.113	0.125
BASS_P05	0.096	0.636	0.628	1.581	0.133	0.155
BASS_P06	-0.005	0.312	0.297	1.567	0.069	0.072
BASS_P07	0.013	0.191	0.180	2.684	0.092	0.097
BASS_P08	0.095	0.403	0.393	2.120	0.088	0.095
BASS_P09	0.075	0.391	0.380	1.763	0.120	0.139
BASS_P11	-0.081	0.474	0.464	2.052	0.104	0.109
BASS_P12	0.182	0.435	0.436	1.748	0.113	0.120
BASS_P13	0.151	0.612	0.615	2.228	0.104	0.119
BASS_P14	0.041	0.540	0.536	1.458	0.126	0.154
BASS_P15	0.093	0.386	0.389	1.777	0.101	0.115
BASS_P17	0.095	0.428	0.421	1.773	0.088	0.096
BASS_P18	0.090	0.516	0.496	2.391	0.083	0.086
BASS_P19	0.068	0.537	0.501	2.431	0.068	0.078
BASS_P20	0.067	0.326	0.319	2.407	0.096	0.098
BASS_P21	0.075	0.401	0.340	2.607	0.092	0.108
BASS_P22	-0.029	0.322	0.300	2.149	0.087	0.090
BASS_P23	0.267	0.541	0.525	2.166	0.054	0.061
BASS_P24	0.106	0.350	0.302	1.626	0.048	0.052

Table S3: Per-gene results and metric definitions. The accompanying digital table, `source_data/SuppTable3_per_gene.csv.gz`, contains the exported Wu and Bassiouni gene-level results, with run, patient, panel, mask and method identifiers. Compression changes storage only. Every metric is calculated using the hidden locations scored by the corresponding protocol. The definitions below explain the gene-level statistics and their summaries; genes and repeated runs are not independent patient observations.

Quantity	Definition and interpretation
Pearson correlation	Correlation between predicted and measured values across scored locations, calculated separately for each gene. Constant targets or predictions yield undefined correlations.
Spearman correlation	Correlation between the ranks of predicted and measured values across scored locations, calculated separately for each gene. Numerical changes in tied ranks do not establish a biological improvement.
Mean squared error (MSE)	Mean squared prediction error across scored locations for each gene, evaluated in the common development-standardized log1p-count units.
Mean absolute error (MAE)	Mean absolute prediction error across scored locations for each gene, evaluated in the same target units.
Panel correlation	Median of defined gene correlations. Undefined correlations are excluded from this median and do not count as positive-correlation successes.
Panel error	Arithmetic mean of the gene-level MSEs or MAEs. All genes are retained in error summaries.
Patient and cohort summaries	Mask seeds are averaged within geometry and geometries within patient. Headline cohort values are medians of patient summaries; paired effects are calculated within patients before summarizing them.

Table S4: Negative-control families. Control definitions are experiment-specific and are given in the corresponding figure legends. Training-row controls differ from inference-time controls that permute permitted sparse inputs. Permuting anchor-expression assignments and anchor-position assignments disrupts related associations and does not constitute independent biological evidence. Exported control results are supplied in `source_data/SuppTable4_negative_controls.csv`.

Control family	Interpretation
Training-target or training-embedding row permutation	Disrupts local morphology–target pairing during development while preserving the validation and evaluation data.
Anchor-expression assignment permutation	Reassigns observed expression vectors among anchors, disrupting their correspondence with spatial position.
Anchor-position assignment permutation	Disrupts the spatial assignment of observed anchors while retaining the permitted molecular information.
Image-location or image-prediction permutation	Tests dependence on the image input while retaining the allowed molecular measurements.

Table S5: Recorded integrity and replay checks. Counts are taken from the original Wu and Bassiouni technical audits; these audits were not rerun during manuscript preparation. Method predictions and metric arrays are distinct audit units. For each prediction, hidden labels were independently replaced by NaNs and random values to test that unobserved expression could not influence model inputs. Digital records: `source_data/SuppTable5_integrity.csv`.

Audit quantity	Wu	Bassiouni
Mask trajectories	132	968
Clean run bundles	396	2,904
Control run bundles	18	18
Method prediction replays	1,656	8,766
Metric array replays	6,624	35,064
Hidden-label corruption checks	3,312	17,532

Table S6: Observation masks and sampling protocols. Coverage is the proportion of retained locations with observed complete expression vectors, not the proportion of tissue area assayed. All unobserved labels are excluded from model inputs. The three regional geometries and ten random-mask seeds are computational replicates. The accompanying `source_data/SuppTable6_masks.csv` records the exported mask identities, patients, regimes, geometries and sampling trajectories.

Protocol component	Definition
Measurement budgets	For N retained locations and $b \in \{0.10, 0.20, 0.40\}$, the observed count is $\lfloor bN + 0.5 \rfloor$. Random anchor sets use nested prefixes of the same trajectory as coverage increases.
Distributed interpolation (V1)	Anchors may be sampled throughout the retained tissue; every remaining location is scored. Predictions use up to 16 nearest observed anchors with normalized inverse-distance-squared weights.
Contiguous regional completion (V2)	A prespecified contiguous region contains the 40% of locations nearest to its center. Anchors are sampled outside that region, and only locations within it are scored. Predictions use the mean of up to 16 nearest observed anchors.
Regional geometry	Three centers are selected at the 25th, 50th and 75th percentile of first coordinate principal-component rank. The three geometries are averaged equally within patient.
Secondary spatial sampling	Label-free farthest-point sampling starts from the eligible location farthest from the eligible-coordinate centroid. It is not optimized using external expression outcomes.
Development masks	Earlier block and capture-window masks differ from the external regional-completion protocol and are interpreted separately.

Table S7: Distance-resolved evaluation and patient support. Distance is measured from each scored hidden location to its nearest observed anchor. Empty bins are omitted. Bassiouni binwise Pearson correlations require at least 20 scored locations in the corresponding run; Wu distance bins report MSE and MAE only. Error summaries may include smaller occupied bins. The patient counts below summarize the Bassiouni cohort at 20% coverage and do not define a universal distance cutoff. Exported bin-level metrics and support counts are supplied in `source_data/SuppTable7_distance_bins.csv`.

Evaluation component	Definition
Distance bins (μm)	$[0, 150)$, $[150, 300)$, $[300, 600)$, $[600, 1200)$, $[1200, 2400)$ and $[2400, \infty)$.
Bin support	Number of scored hidden locations contributing to the patient, method, mask and distance-bin summary; occupied-bin support may differ across patients and protocols.
Nearest-to-farthest comparison	Compares MSE in the nearest and farthest occupied bins within each patient.
Monotonicity	Assesses whether MSE strictly increases across the occupied distance bins.

Bassiouni patient summary	V1	V2
Higher MSE in farthest than nearest occupied bin	15/22	20/22
Strictly increasing occupied-bin MSE	6/22	13/22

Table S8: Selected supporting predictor architectures for the 50-gene panel. Dimensions denote input → hidden → output, and every listed MLP has GELU, dropout 0.1 and a linear output layer. Encoder parameters remain frozen; counts include trainable head biases. All MLPs use the shared optimizer and stopping rule above. The initial local predictor and the representation-comparison predictor are distinct fitted heads.

Predictor	Selected dimensions	Fitting settings or fixed aggregation
Initial ResNet18 ridge	$512 \to 50$	Ridge penalty 100; intercept; 25,650 coefficients.
Initial ResNet18 MLP	$512 \to 128 \to 50$	Lowest validation MSE; 72,114 head parameters.
ResNet18 mean pooling / GCN	$512 \to 128 \to 50$	PX or QX on capped Delaunay adjacency; 72,114 head parameters.
ResNet18 mean-GraphSAGE	$1024 \to 67 \to 50$	$[X, RX]$ on capped Delaunay adjacency; 72,075 head parameters.
ResNet18 representation-comparison MLP	$512 \to 256 \to 50$	Highest validation median gene Pearson; 144,178 head parameters.
Phikon local MLP	$768 \to 256 \to 50$	Same candidate grid and selection criterion; 209,714 head parameters.
Phikon spatial MLPs	$768 \to 128 \to 50$; $1536 \to 67 \to 50$	Mean pooling and mean-GraphSAGE, respectively; inherited graph and head widths.
Sparse residual correction	34 covariates plus intercept per gene	Gene-specific ridge correction of fixed fusion; penalty 10,000; 1,750 coefficients for 50 genes.

Table S9: Coverage response and contiguous-region completion in the two original evaluation studies. All rows use the same frozen 50-gene panel and spatial interpolation. Values are medians of patient summaries, with $n = 3$ for Wu and $n = 22$ for Bassiouni. V1 is distributed interpolation with inverse-distance-squared weighting; V2 is completion of a predefined contiguous region using the mean of nearby anchors. Both the scoring support and interpolation rule change between V1 and V2. The comparison therefore describes protocol-level deterioration rather than an isolated causal geometry effect. In both studies, every evaluation patient improved in Pearson correlation and MSE from 10% to 40% V1 coverage; every patient deteriorated in Pearson, MSE and MAE under V2 relative to V1 at 20%.

Cohort	Coverage	V1 Pearson r	V1 MSE	V2 Pearson r	V2 MSE
Wu, original evaluation	10%	0.313	0.1080	0.111	0.1356
	20%	0.359	0.0987	0.081	0.1366
	40%	0.393	0.0899	0.088	0.1385
Bassiouni, original evaluation	10%	0.380	0.1001	0.121	0.1520
	20%	0.416	0.0923	0.171	0.1496
	40%	0.465	0.0852	0.181	0.1518

S4 Additional controls and source-data guide

The matched molecular and neural comparisons use previously evaluated cohorts. Complete source tables are under `source_data/journal/` and `source_data/extension/`; protocols, input identities, fitted-state records and numerical audits are documented in the repository. Exact Bassiouni inputs are available and verified. The complete additional per-gene exports and model states remain author-local with release-ready manifests, not publicly deposited assets. Publication regeneration from exported tables is distinct from replay of the three missing historical primary image heads.

Table S10: All 20 matched-information estimators in Wu at 20% V1 coverage. Each patient value averages ten original random trajectories. P1=1160920F, P2=CID4290, P3=CID44971. Relative MSE is candidate minus fixed IDW k16/p2, divided by that reference; negative values improve. No external winner was selected. All 10/20/40% results, MAE and individual paired effects accompany the CSV exports. Gaussian bandwidth uses observed-anchor geometry only.

Estimator	r P1	r P2	r P3	MSE% P1	MSE% P2	MSE% P3
nearest	0.223	0.228	0.368	+68.3	+61.9	+63.7
uniform_k4	0.328	0.293	0.474	+8.0	+8.2	+7.5
uniform_k8	0.354	0.294	0.475	+0.7	+4.6	+4.0
uniform_k16	0.359	0.265	0.442	-0.8	+7.5	+6.8
uniform_k32	0.348	0.220	0.396	+0.8	+13.7	+11.7
idw_k4_p1	0.327	0.305	0.483	+8.5	+6.6	+6.4
idw_k4_p2	0.313	0.299	0.475	+13.1	+9.7	+10.2
idw_k4_p3	0.297	0.289	0.460	+19.8	+15.3	+16.3
idw_k8_p1	0.360	0.321	0.499	-0.0	+0.8	+0.3
idw_k8_p2	0.341	0.317	0.495	+4.4	+2.8	+3.0
idw_k8_p3	0.315	0.301	0.474	+12.7	+9.4	+10.4
idw_k16_p1	0.378	0.318	0.497	-3.1	+0.7	+0.0
idw_k16_p2	0.359	0.325	0.502	+0.0	+0.0	+0.0
idw_k16_p3	0.324	0.305	0.479	+9.1	+6.6	+7.6
idw_k32_p1	0.387	0.307	0.480	-3.3	+3.8	+2.1
idw_k32_p2	0.371	0.328	0.503	-2.2	-0.8	-1.2
idw_k32_p3	0.329	0.308	0.481	+7.2	+5.2	+6.3
gaussian_k8	0.360	0.308	0.489	-0.1	+2.6	+2.0
gaussian_k16	0.374	0.286	0.461	-1.9	+4.8	+4.4
gaussian_k32	0.358	0.249	0.417	-0.5	+10.3	+9.5

Table S11: Operator by historical protocol at 20% coverage. IDW and uniform operators share each original mask and scored set. V2 averages geometries equally after averaging seeds. The V1/V2 contrast still changes spatial support and is not a causal geometry estimate. All budgets are supplied in source data.

Patient	Protocol	IDW r	IDW MSE	Uniform r	Uniform MSE
1160920F	V1	0.359	0.0817	0.359	0.0811
1160920F	V2	0.179	0.1029	0.176	0.1028
CID4290	V1	0.325	0.0987	0.265	0.1061
CID4290	V2	0.050	0.1367	0.036	0.1366
CID44971	V1	0.502	0.1150	0.442	0.1228
CID44971	V2	0.113	0.1781	0.081	0.1767

Table S12: Identical-target anchor-placement effects at 20% coverage in Wu. Region-exclusion minus distributed anchors, with the same H and total anchor count. Each row averages ten seeds. The holdout H is fixed across conditions and budgets. All three patients deteriorate in Pearson, MSE and MAE with both estimators. This central-region result is exploratory robustness evidence in previously inspected patients, not new blind validation.

Patient	Operator	Δr	Relative MSE%	Δ MAE
1160920F	IDW	-0.1056	+20.1	+0.0214
1160920F	Uniform	-0.0928	+21.7	+0.0196
CID4290	IDW	-0.2452	+45.4	+0.0464
CID4290	Uniform	-0.1875	+36.0	+0.0361
CID44971	IDW	-0.1274	+37.3	+0.0386
CID44971	Uniform	-0.1605	+38.9	+0.0387

Table S13: All new controlled neural fits at 20% TENX99 coverage. Each value averages three identical original masks. Models fit NCBI783/NCBI785 and use TENX95 alone for stopping. Seeds are training variability, not patients. Residual is the exploratory image plus interpolated observed-residual predictor. All budgets, fixed 20% fusion and anchor-only values are in CSVs. No architecture or seed was selected using TENX99. Nominal parameter counts are 72,114 (local/mean/GCN) and 72,075 (SAGE); SAGE self-only duplicates self inputs and has a smaller effective function class.

Configuration	Seed	Image r	Image MSE	Residual r	Residual MSE
local local	0	0.666	0.9142	0.818	0.2576
local local	1	0.686	0.9008	0.824	0.2569
local local	2	0.695	0.9191	0.821	0.2555
MEAN spatial	0	0.651	0.8825	0.812	0.2458
MEAN spatial	1	0.647	0.9095	0.813	0.2497
MEAN spatial	2	0.658	0.9206	0.813	0.2446
MEAN self	0	0.666	0.9142	0.818	0.2576
MEAN self	1	0.686	0.9008	0.824	0.2569
MEAN self	2	0.695	0.9191	0.821	0.2555
MEAN shuffled	0	0.117	1.2311	0.787	0.2747
MEAN shuffled	1	0.227	1.1425	0.771	0.2865
MEAN shuffled	2	0.213	1.1819	0.778	0.2889
GCN spatial	0	0.652	0.8871	0.809	0.2508
GCN spatial	1	0.659	0.8885	0.812	0.2518
GCN spatial	2	0.653	0.9240	0.809	0.2503
GCN self	0	0.666	0.9142	0.818	0.2576
GCN self	1	0.686	0.9008	0.824	0.2569
GCN self	2	0.695	0.9191	0.821	0.2555
GCN shuffled	0	0.124	1.2297	0.788	0.2743
GCN shuffled	1	0.230	1.1264	0.767	0.2909
GCN shuffled	2	0.228	1.1830	0.777	0.2908
SAGE spatial	0	0.667	0.8611	0.823	0.2340
SAGE spatial	1	0.688	0.8792	0.831	0.2346
SAGE spatial	2	0.679	0.9173	0.822	0.2329
SAGE self	0	0.667	0.9364	0.815	0.2574
SAGE self	1	0.656	0.9010	0.812	0.2590
SAGE self	2	0.672	0.9599	0.810	0.2582
SAGE shuffled	0	0.686	0.8615	0.812	0.2511
SAGE shuffled	1	0.695	0.8655	0.816	0.2564
SAGE shuffled	2	0.684	0.9124	0.818	0.2403

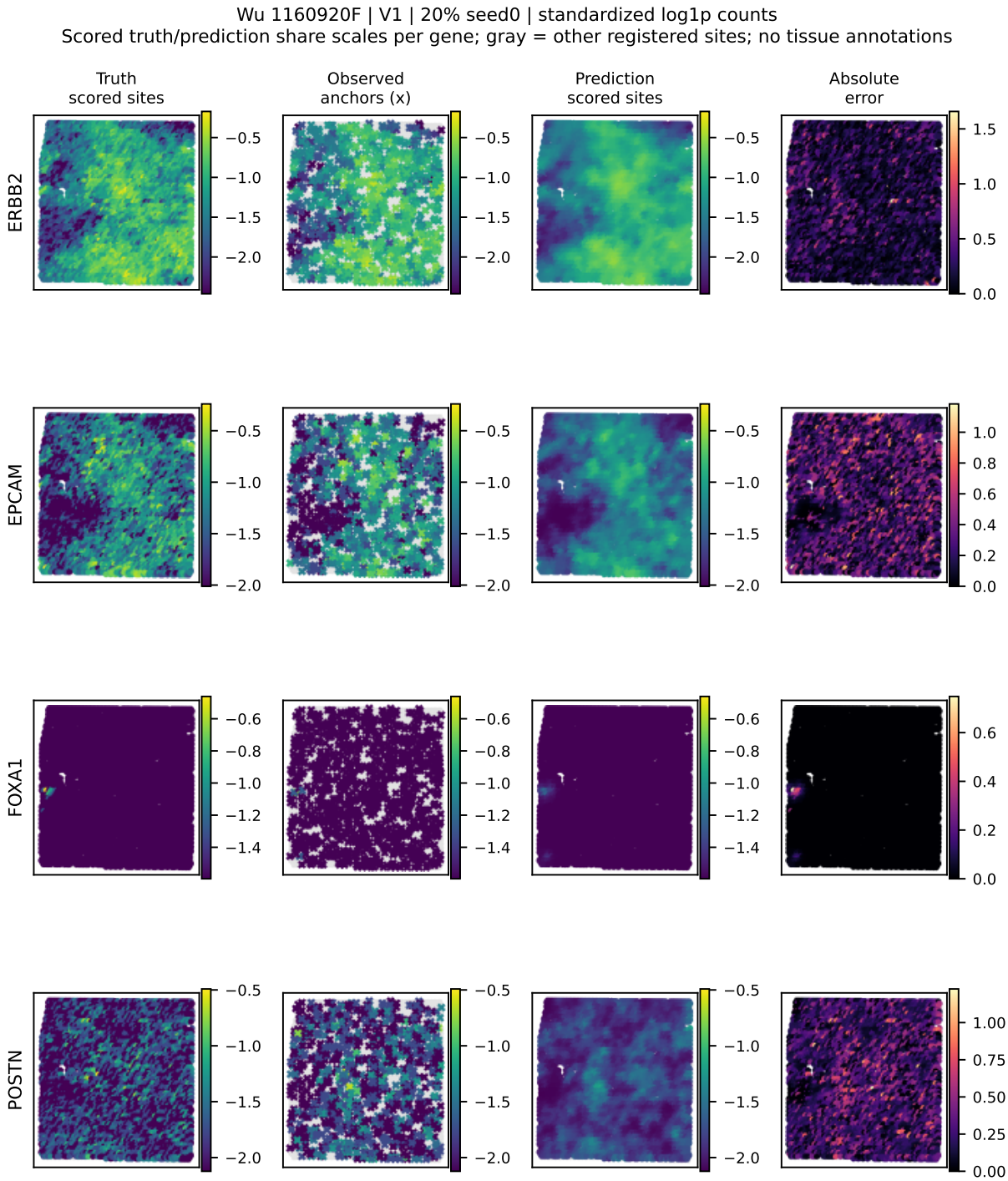


Figure S13: Predetermined real Wu V1 gene maps. Patient 1160920F, 20% random seed0, original mask, fixed IDW k16/p2. Genes ERBB2, EPCAM, FOXA1 and POSTN were designated before comparisons. Truth/prediction use shared per-gene scales over scored sites; errors have separate scales. Anchors are crosses and gray sites are other registered locations. Standardized log1p counts are not absolute transcript abundance. Constant scored fields are labelled rather than omitted or given misleading contrast. Source: Wu et al., Zenodo 4739739, CC-BY4.0; no tumor/cell annotations are inferred.

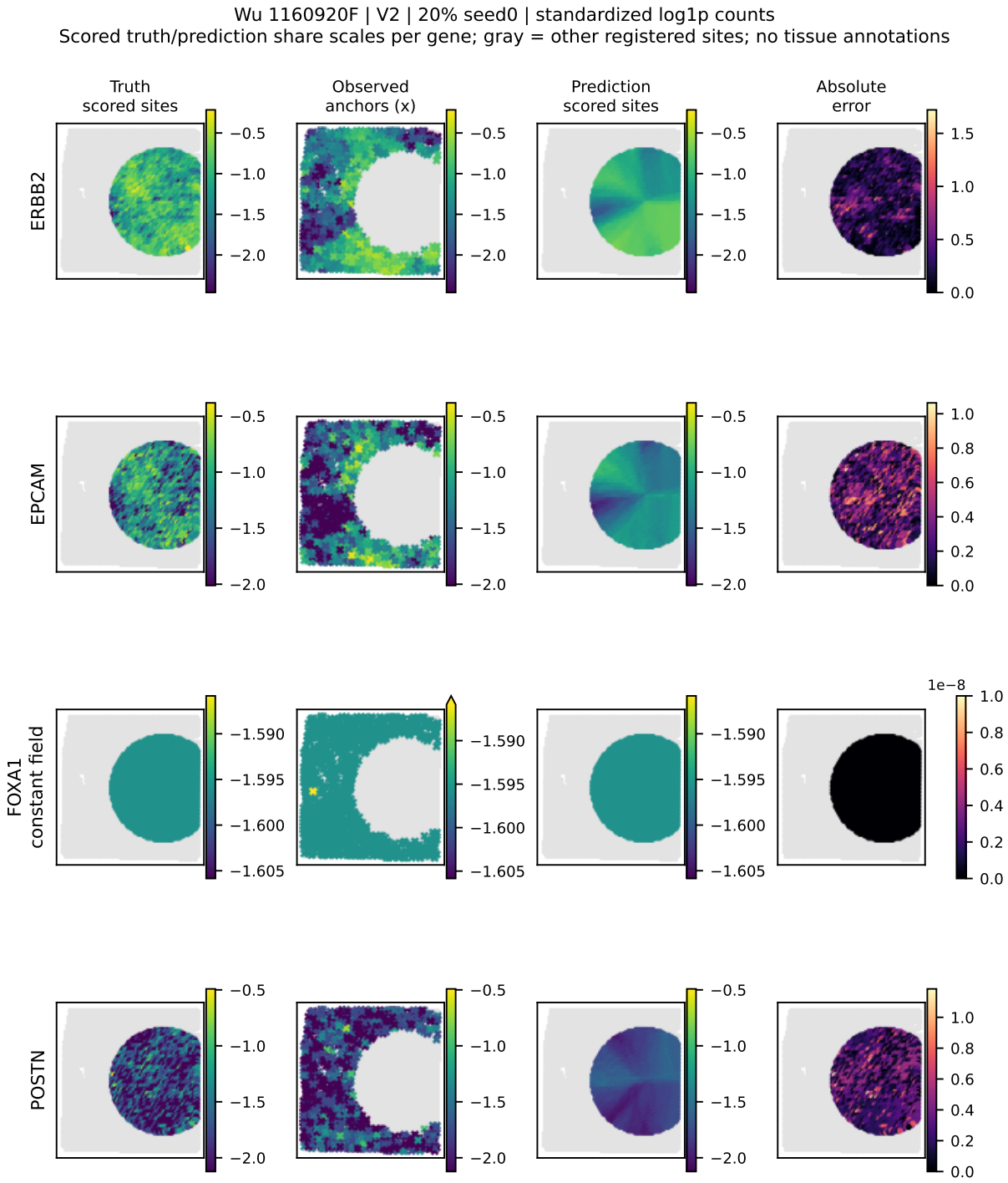


Figure S14: Predetermined real Wu V2 gene maps. Same patient, genes, 20% budget and seed0, central original V2 geometry 1, historical uniform k16 operator. Colored truth/prediction/error points are the original forbidden scoring region; anchor crosses lie outside. Scales and source terms match the preceding figure. FOXA1 has a constant scored field; numerical residuals do not establish biological recovery. These maps retain unsuccessful/constant examples and are not independent validation. The corresponding Bassiouni maps are provided in Supplementary Figs. S26 and S27.

S5 Graph comparisons and computational diagnostics

S5.1 Complete cohort and prediction-route matrices

All matrices use 20% distributed coverage and the fixed 50-gene panel. Sixteen full-section configurations and ten frozen minibatch configuration families are shown, each retaining three training seeds, alongside fixed IDW and uniform molecular estimators. Metrics are averaged over the same three mask seeds and then training seeds within patient. Dots are patient summaries, and black ticks are cohort medians; their spread is not a confidence interval. The image-only route uses frozen image features and permitted coordinates; fixed fusion and observed-residual interpolation additionally use the same measured anchors as the molecular references. Thus image-only versus molecular rows have different information access. Architecture names identify configurations, not outcome-selected winners.

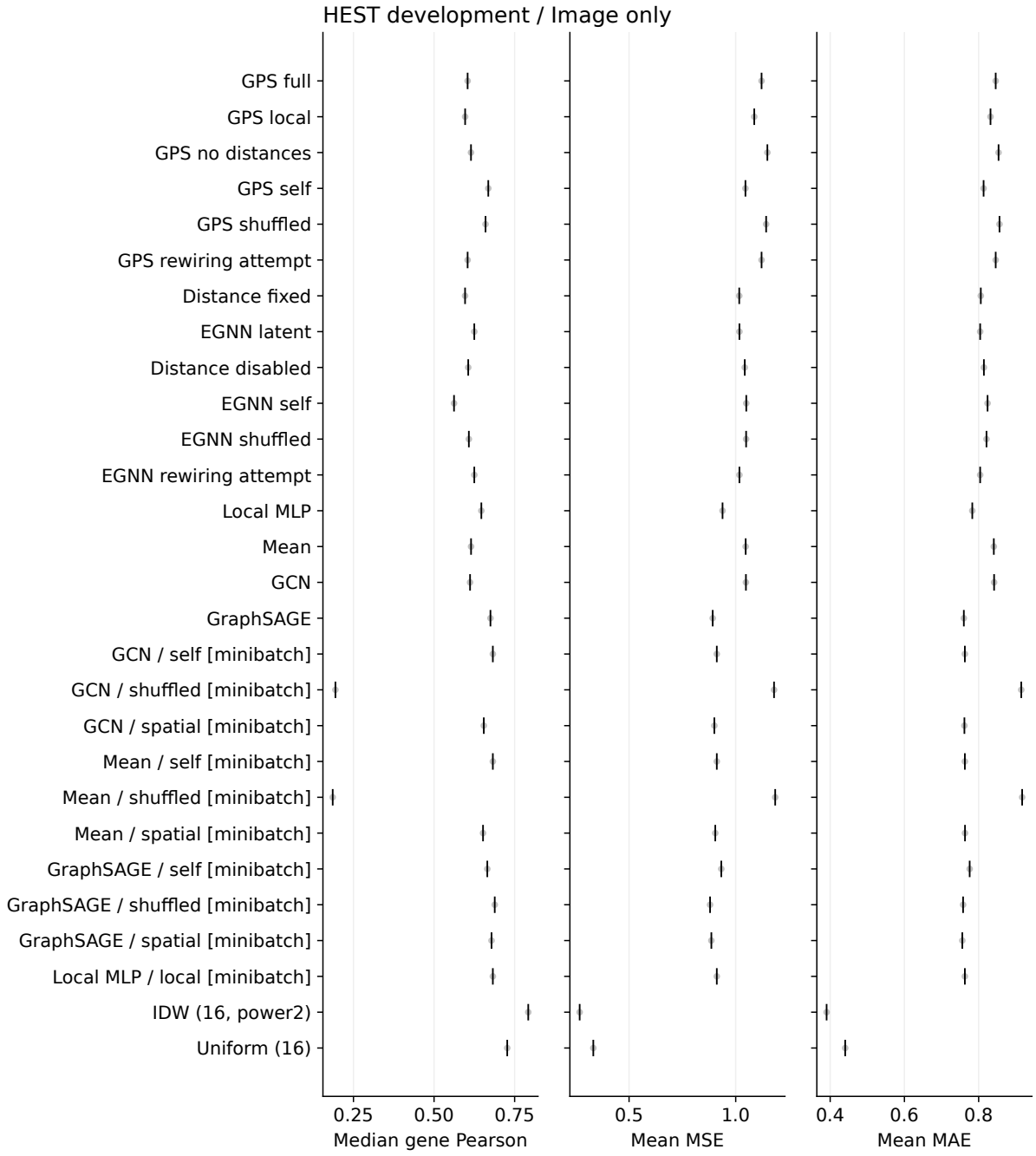


Figure S15: HEST development: image-only at 20% distributed coverage. Results for one previously evaluated patient, TENX99. Pearson, MSE and MAE are shown after averaging three mask and three training seeds within patient. Neural rows use image features and permitted geometry; the two molecular rows receive measured anchors. The dot represents one patient; no population interval is inferred.

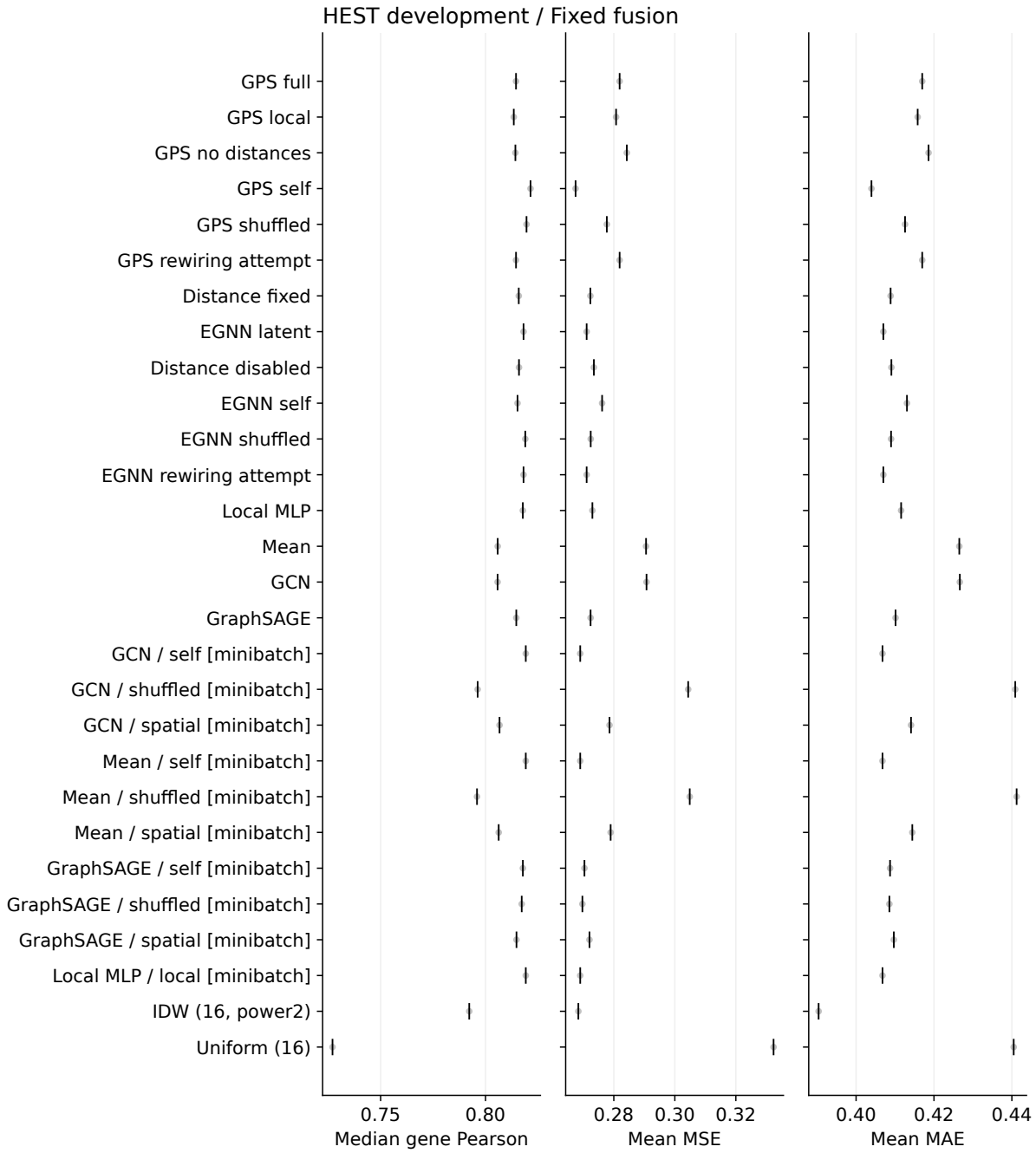


Figure S16: HEST development: fixed fusion at 20% distributed coverage. Results for one previously evaluated patient, TENX99. Pearson, MSE and MAE are shown after averaging three mask and three training seeds within patient. Neural rows combine 0.2 calibrated image prediction with 0.8 IDW; molecular rows use the same anchor vectors without images. The dot represents one patient; no population interval is inferred.

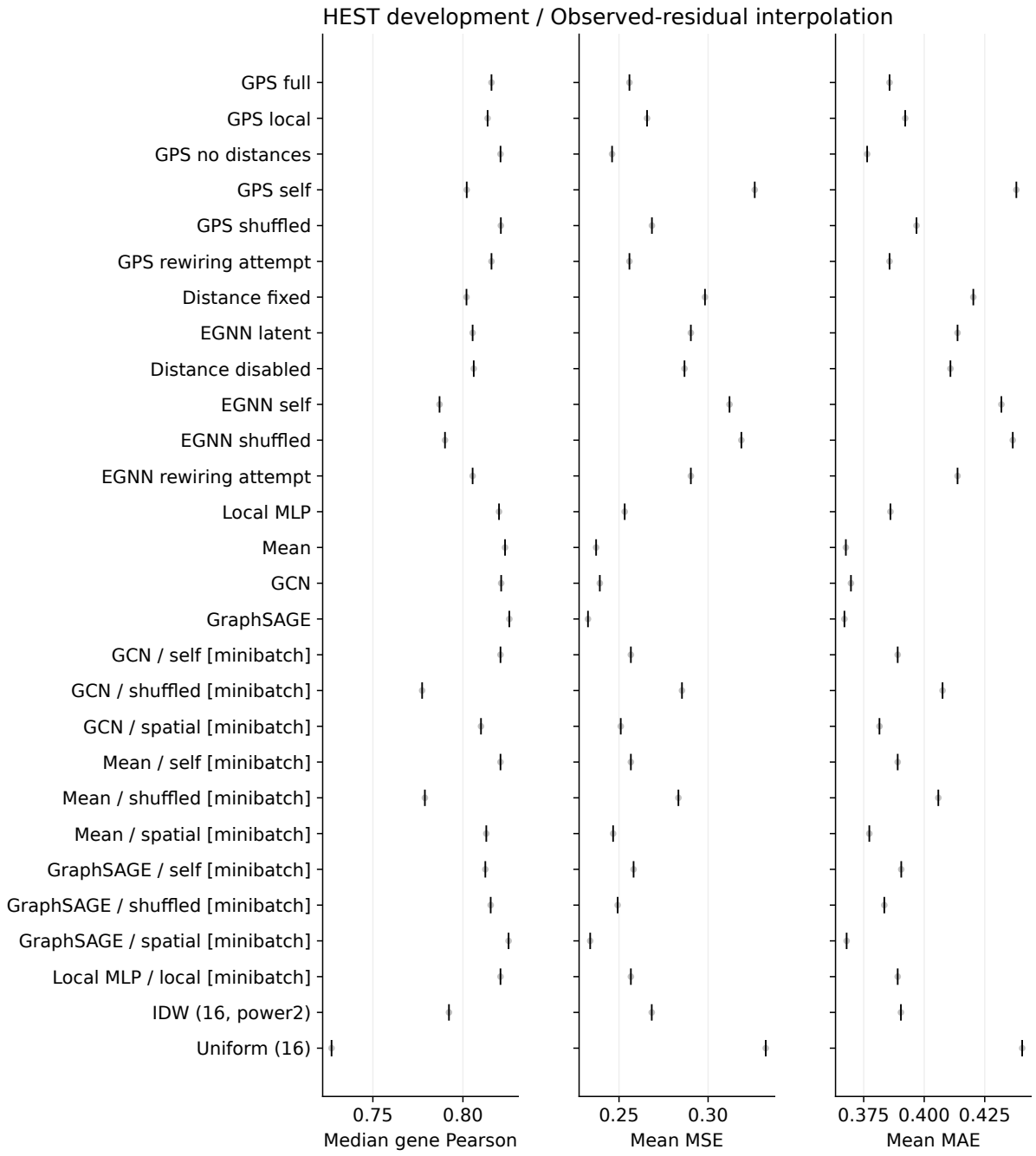


Figure S17: HEST development: observed-residual interpolation at 20% distributed coverage. Results for one previously evaluated patient, TENX99. Pearson, MSE and MAE are shown after averaging three mask and three training seeds within patient. Neural rows add interpolation of observed residuals to image prediction; molecular rows use the same anchors without images. The dot represents one patient; no population interval is inferred.

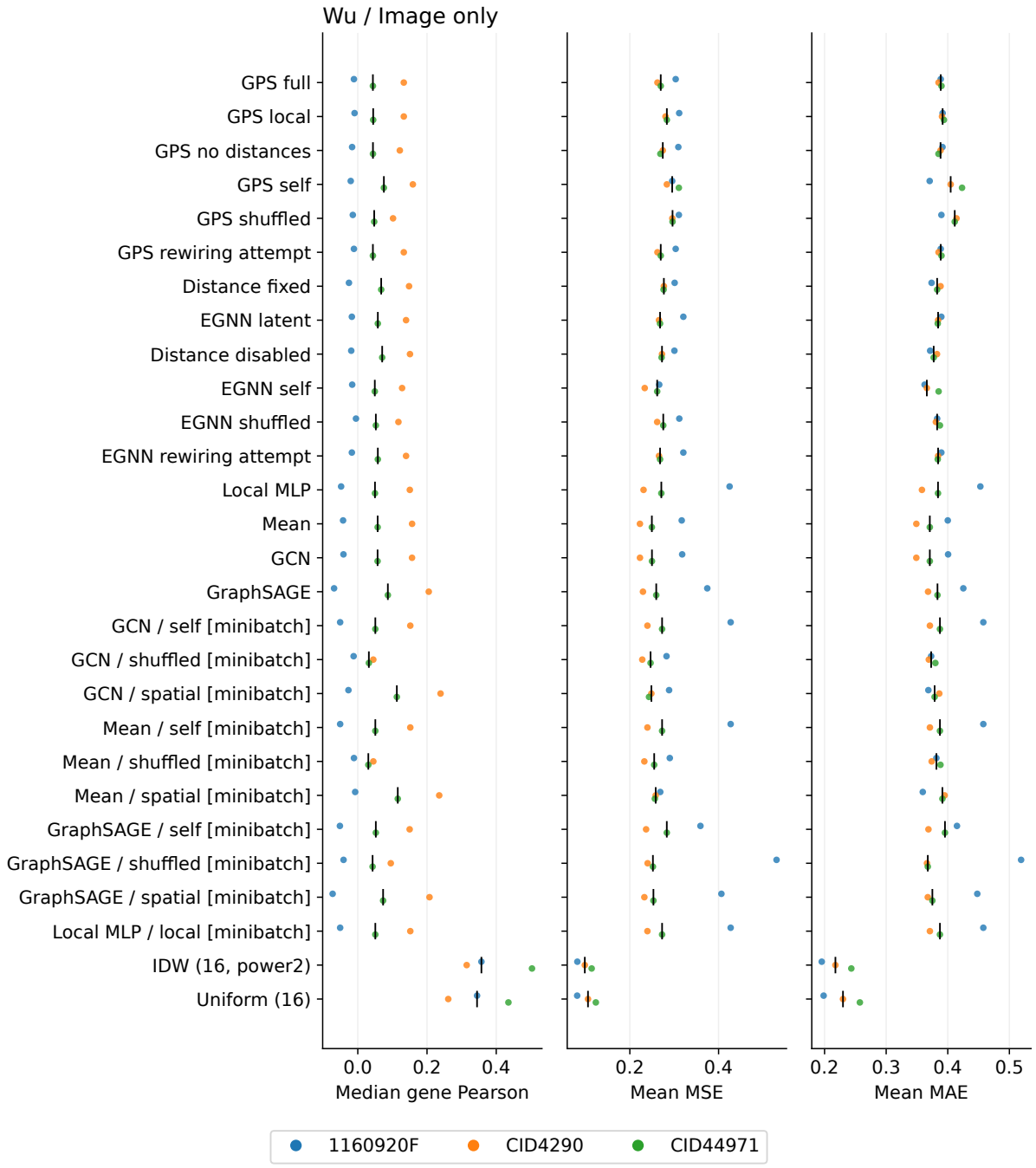


Figure S18: Wu: image-only at 20% distributed coverage. Results for three previously evaluated patients. Pearson, MSE and MAE are shown after averaging three mask and three training seeds within patient. Neural rows use image features and permitted geometry; the two molecular rows receive measured anchors. Colors identify 1160920F, CID4290 and CID44971 individually.

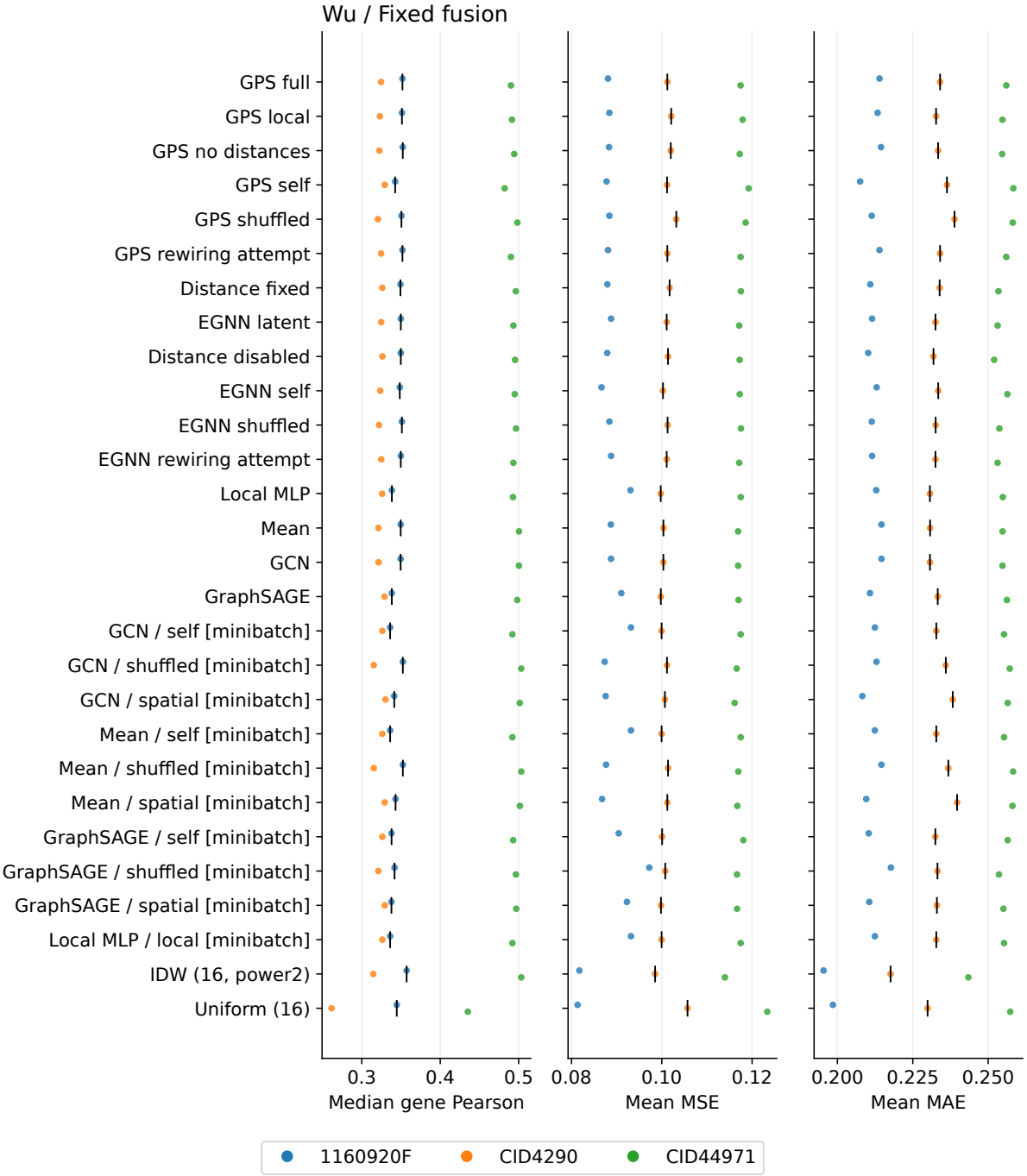


Figure S19: Wu: fixed fusion at 20% distributed coverage. Results for three previously evaluated patients. Pearson, MSE and MAE are shown after averaging three mask and three training seeds within patient. Neural rows combine 0.2 calibrated image prediction with 0.8 IDW; molecular rows use the same anchor vectors without images. Colors identify 1160920F, CID4290 and CID44971 individually.

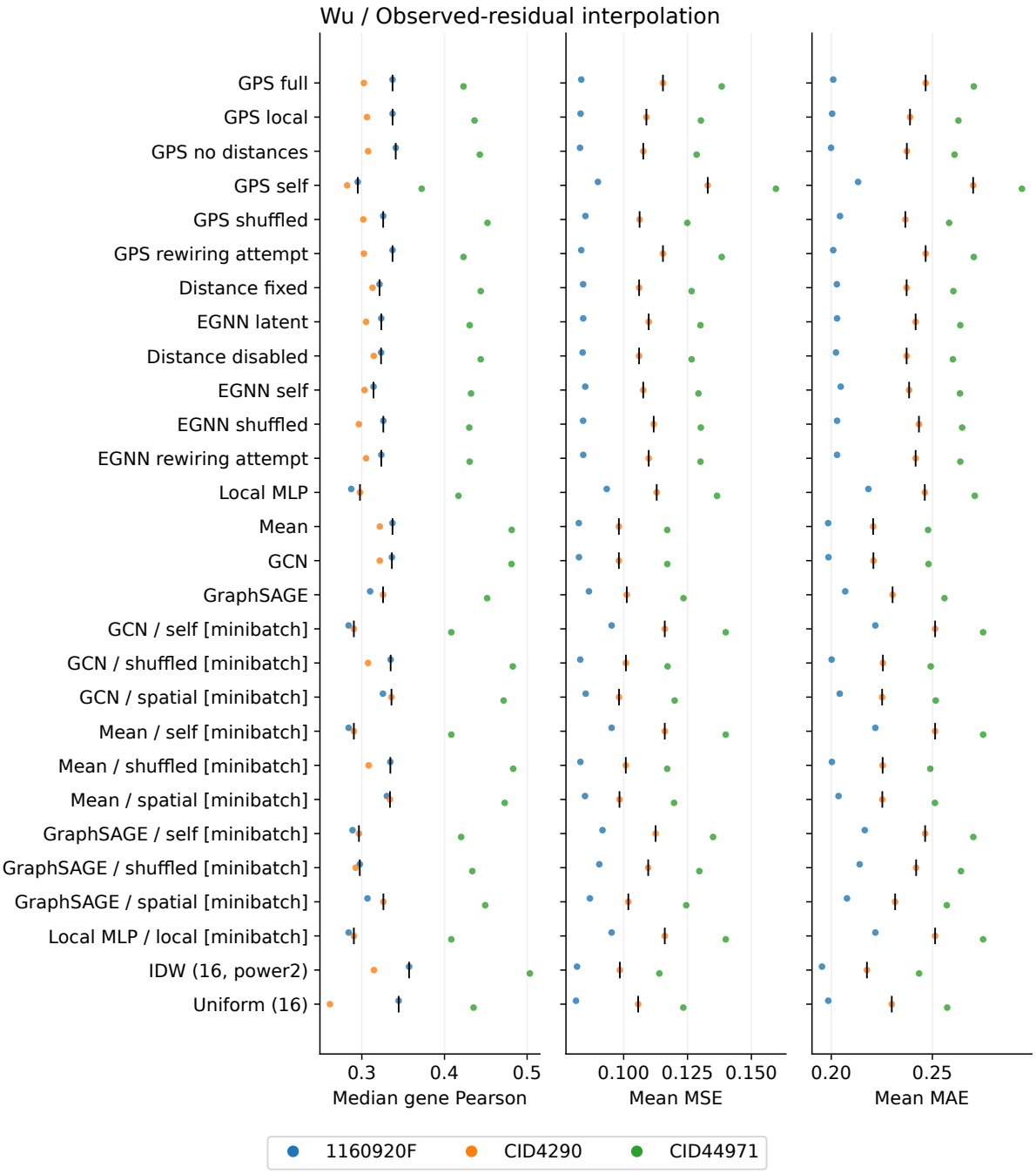


Figure S20: Wu: observed-residual interpolation at 20% distributed coverage. Results for three previously evaluated patients. Pearson, MSE and MAE are shown after averaging three mask and three training seeds within patient. Neural rows add interpolation of observed residuals to image prediction; molecular rows use the same anchors without images. Colors identify 1160920F, CID4290 and CID44971 individually.

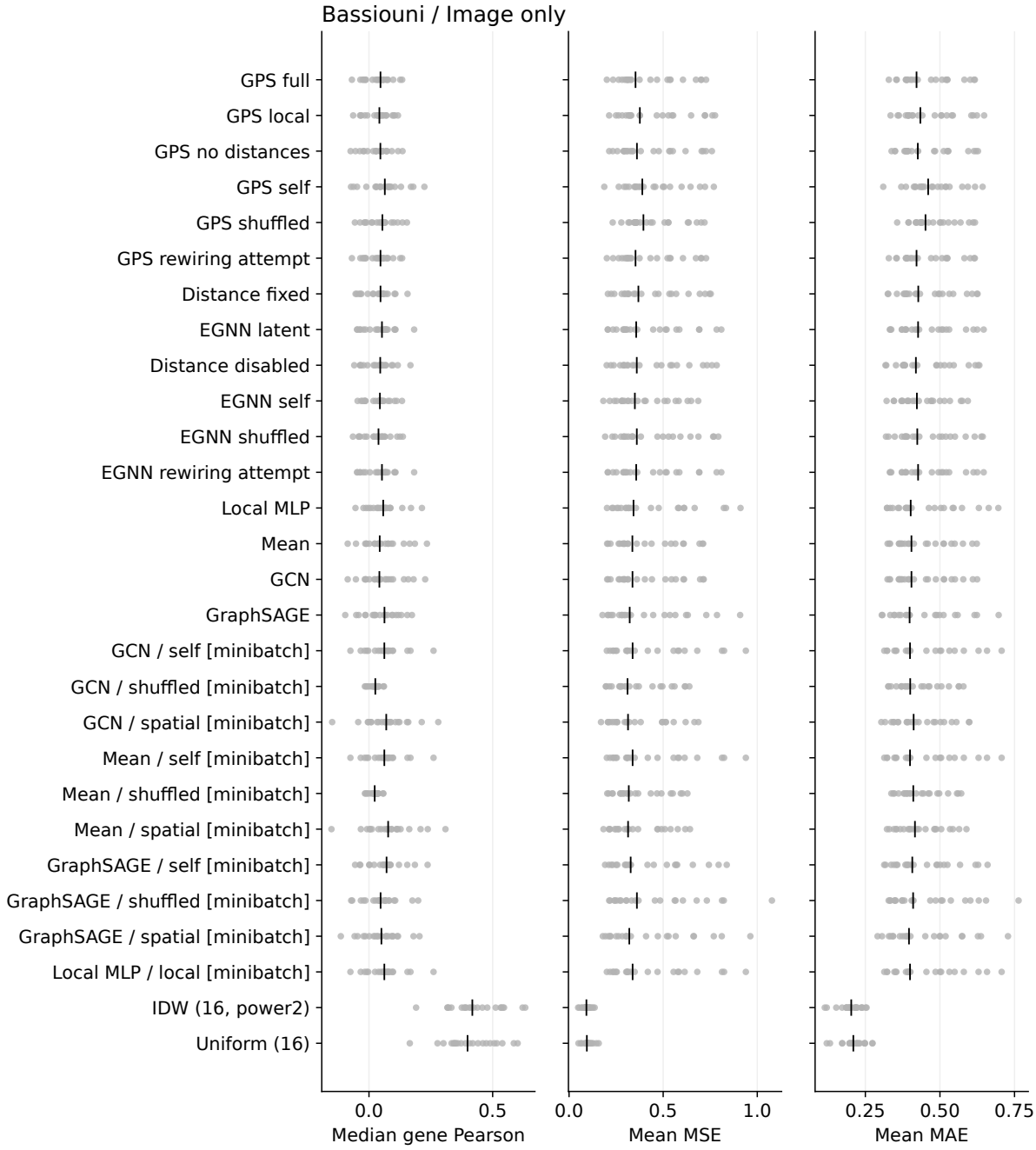


Figure S21: Bassiouni: image-only at 20% distributed coverage. Results for 22 previously evaluated patients. Pearson, MSE and MAE are shown after averaging three mask and three training seeds within patient. Neural rows use image features and permitted geometry; the two molecular rows receive measured anchors. Gray dots identify individual patients; black ticks are medians of their summaries.

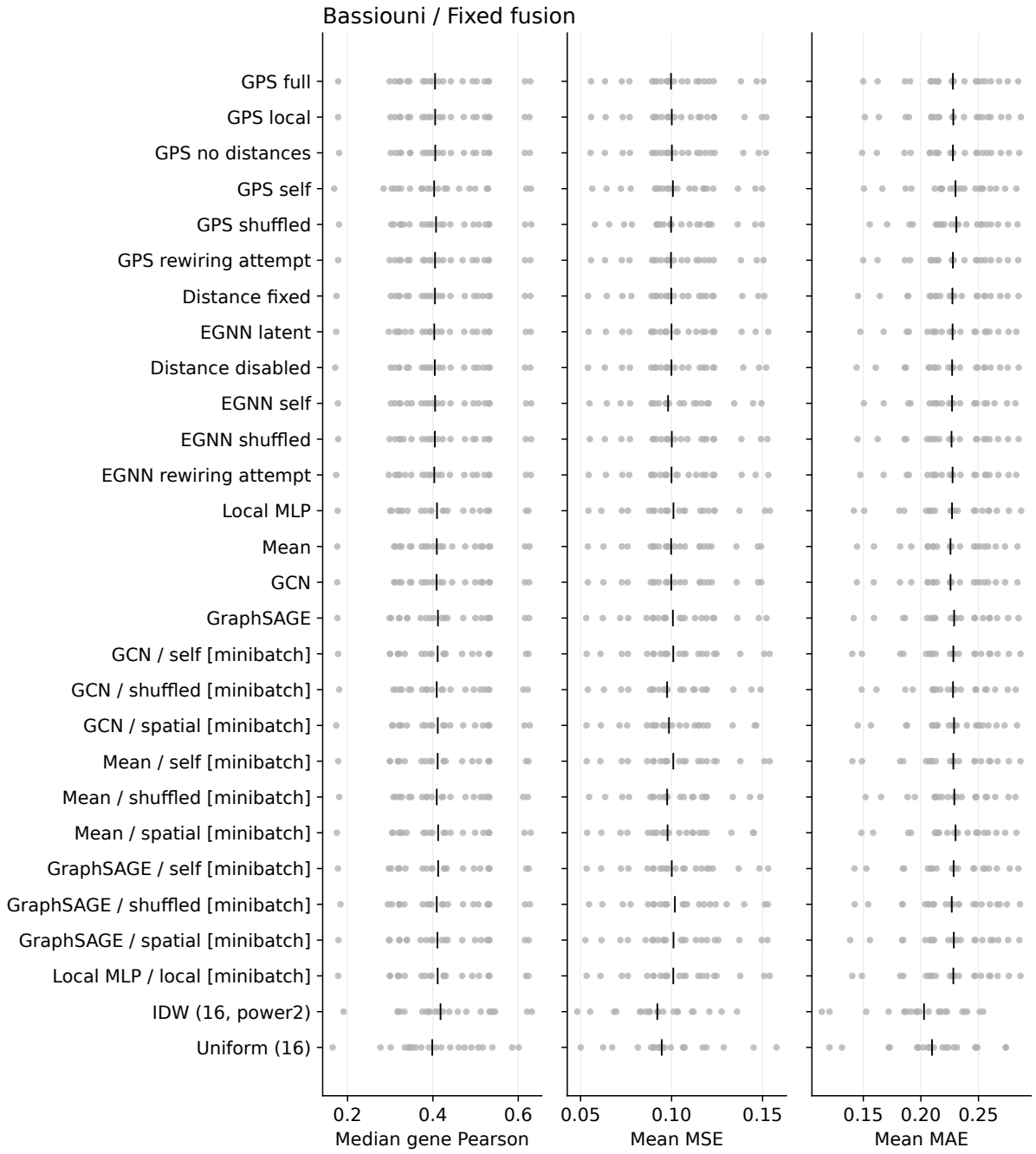


Figure S22: Bassiouni: fixed fusion at 20% distributed coverage. Results for 22 previously evaluated patients. Pearson, MSE and MAE are shown after averaging three mask and three training seeds within patient. Neural rows combine 0.2 calibrated image prediction with 0.8 IDW; molecular rows use the same anchor vectors without images. Gray dots identify individual patients; black ticks are medians of their summaries.

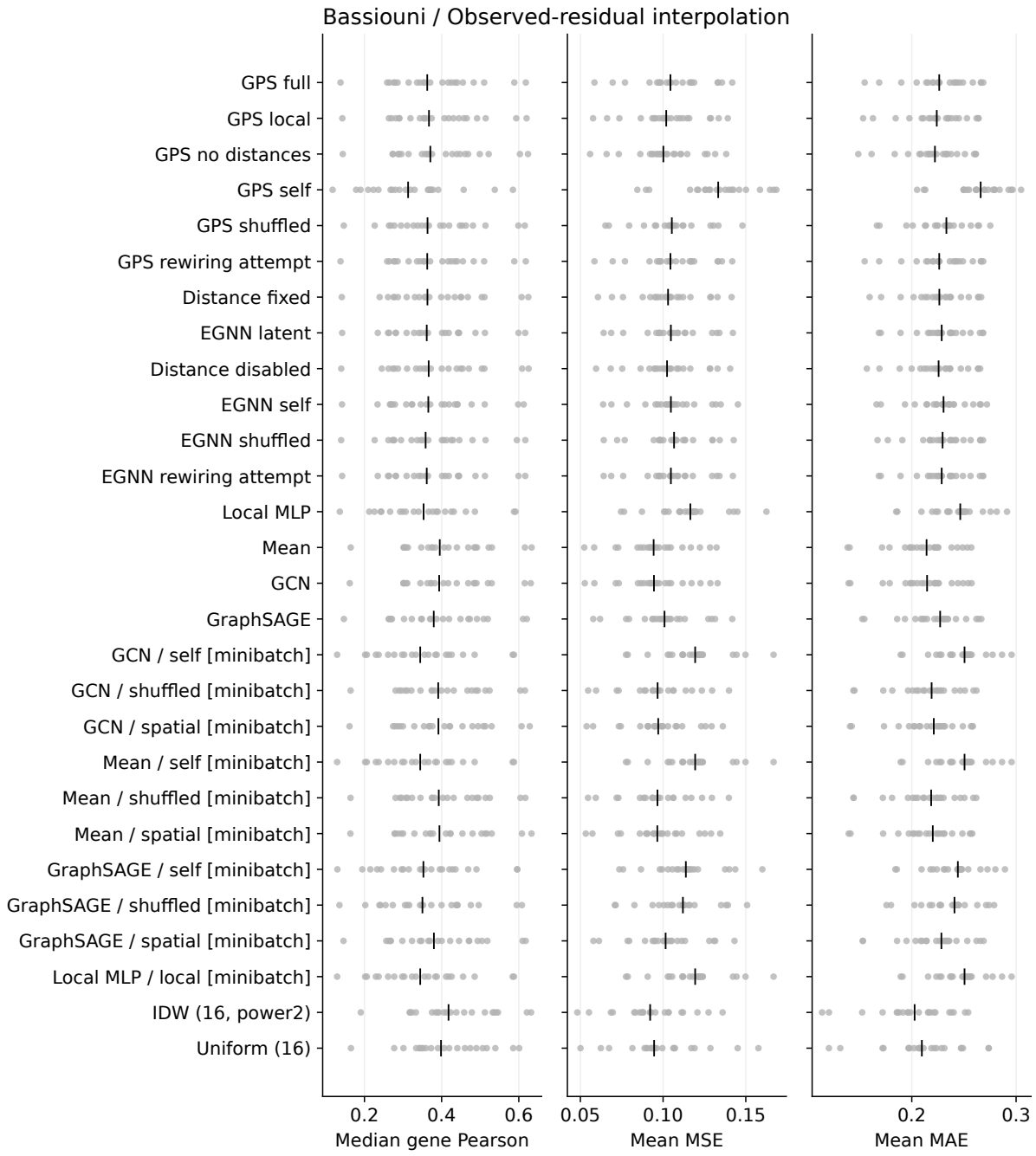


Figure S23: Bassiouni: observed-residual interpolation at 20% distributed coverage. Results for 22 previously evaluated patients. Pearson, MSE and MAE are shown after averaging three mask and three training seeds within patient. Neural rows add interpolation of observed residuals to image prediction; molecular rows use the same anchors without images. Gray dots identify individual patients; black ticks are medians of their summaries.

Table S14: All 384 saved-checkpoint conditional geometric checks. Maxima across all three seeds and recorded transforms. Physical latent errors are in micrometers. Tests are conditional on fixed image features, not the complete CNN pipeline. Baseline preaggregation remains float 32 even when the head is tested in float 64.

Model	Precision	Max scalar error	Max latent error	Checks
Distance disabled	float 64	0	1.14×10^{-13}	12
Distance disabled	float 32	7.75×10^{-7}	2.40×10^{-5}	12
Distance fixed	float 64	1.11×10^{-15}	1.14×10^{-13}	12
Distance fixed	float 32	8.42×10^{-7}	2.40×10^{-5}	12
EGNN rewired	float 64	8.88×10^{-16}	1.14×10^{-13}	12
EGNN rewired	float 32	7.15×10^{-7}	3.13×10^{-5}	12
EGNN latent	float 64	1.33×10^{-15}	1.14×10^{-13}	12
EGNN latent	float 32	7.45×10^{-7}	2.86×10^{-5}	12
EGNN self	float 64	0	1.14×10^{-13}	12
EGNN self	float 32	7.15×10^{-7}	2.40×10^{-5}	12
EGNN shuffled	float 64	1.33×10^{-15}	1.14×10^{-13}	12
EGNN shuffled	float 32	7.45×10^{-7}	2.70×10^{-5}	12
GPS rewired	float 64	1.33×10^{-15}	1.14×10^{-13}	12
GPS rewired	float 32	9.54×10^{-7}	2.40×10^{-5}	12
GPS full	float 64	1.33×10^{-15}	1.14×10^{-13}	12
GPS full	float 32	9.54×10^{-7}	2.40×10^{-5}	12
GPS local	float 64	1.11×10^{-15}	1.14×10^{-13}	12
GPS local	float 32	7.75×10^{-7}	2.40×10^{-5}	12
GPS no distances	float 64	0	1.14×10^{-13}	12
GPS no distances	float 32	8.34×10^{-7}	2.40×10^{-5}	12
GPS self	float 64	0	1.14×10^{-13}	12
GPS self	float 32	7.15×10^{-7}	2.40×10^{-5}	12
GPS shuffled	float 64	1.55×10^{-15}	1.14×10^{-13}	12
GPS shuffled	float 32	9.54×10^{-7}	2.40×10^{-5}	12
GCN	float 64	0	1.14×10^{-13}	12
GCN	float 32	8.94×10^{-8}	2.40×10^{-5}	12
Local MLP	float 64	0	1.14×10^{-13}	12
Local MLP	float 32	3.58×10^{-7}	2.40×10^{-5}	12
Mean	float 64	3.49×10^{-9}	1.14×10^{-13}	12
Mean	float 32	1.19×10^{-7}	2.40×10^{-5}	12
GraphSAGE	float 64	0	1.14×10^{-13}	12
GraphSAGE	float 32	1.19×10^{-7}	2.40×10^{-5}	12

Table S15: Resource accounting for every new configuration. All three seeds are included. Training wall times were measured under the documented concurrent workload and are not a controlled throughput comparison. Standardized inference uses the same 7,760-node TENX95 section, four CPU threads, two warmups and median of five forwards, including model-specific feature aggregation/Tensor conversion but excluding frozen CNN and graph construction. Peak process RSS includes interpreter and data, not only model activations. No GPU or paid external service was used. Full per-seed values, graph construction times and environments are exported.

Model	Parameters	Best epoch range	Fit s	Forward ms	RSS MiB
GPS full	121018	5–60	5.67	78.4	546
GPS local	121018	5–56	7.04	102.3	568
GPS no distances	121018	5–42	5.45	69.6	568
GPS self	121018	47–53	15.76	34.6	568
GPS shuffled	121018	6–76	22.15	76.5	568
GPS rewired	121018	5–60	3.78	81.5	546
Distance fixed	87154	7–53	12.47	56.5	533
EGNN latent	87282	7–57	13.95	54.2	538
Distance disabled	87154	7–74	12.10	54.1	537
EGNN self	87282	8–35	1.91	11.7	534
EGNN shuffled	87282	7–40	14.76	54.6	545
EGNN rewired	87282	7–57	13.95	62.2	544
Local MLP	72114	8–77	5.30	9.9	520
Mean	72114	8–9	1.24	36.2	550
GCN	72114	8–9	1.26	75.8	581
GraphSAGE	72075	39–60	3.77	58.4	550

Table S16: Every new fit, without seed selection. Histories, optimizer states, input hashes and complete checkpoint SHA 256 values are available in the fit JSON files and inventory CSV; weights are distributed separately from Git.

Model	Seed	Best epoch	Ran	Seconds	RSS MiB
Distance disabled	0	74	80	35.82	632
Distance disabled	1	30	40	12.10	643
Distance disabled	2	7	17	5.66	650
Distance fixed	0	53	63	28.79	613
Distance fixed	1	30	40	12.47	638
Distance fixed	2	7	17	4.35	632
EGNN rewired	0	23	33	13.95	642
EGNN rewired	1	57	67	20.21	651
EGNN rewired	2	7	17	5.02	643
EGNN latent	0	23	33	13.95	640
EGNN latent	1	57	67	19.52	639
EGNN latent	2	7	17	5.50	648
EGNN self	0	35	45	4.64	506
EGNN self	1	8	18	1.88	504
EGNN self	2	8	18	1.91	502
EGNN shuffled	0	24	34	18.71	634

Model	Seed	Best epoch	Ran	Seconds	RSS MiB
EGNN shuffled	1	40	50	14.76	635
EGNN shuffled	2	7	17	4.77	643
GPS rewired	0	60	70	35.00	631
GPS rewired	1	5	15	3.70	633
GPS rewired	2	6	16	3.78	624
GPS full	0	60	70	15.56	626
GPS full	1	5	15	5.67	629
GPS full	2	6	16	3.82	629
GPS local	0	56	66	30.11	692
GPS local	1	5	15	5.24	679
GPS local	2	5	15	7.04	674
GPS no distances	0	42	52	31.17	600
GPS no distances	1	5	15	3.67	599
GPS no distances	2	6	16	5.45	596
GPS self	0	51	61	20.15	543
GPS self	1	47	57	15.76	546
GPS self	2	53	63	13.54	546
GPS shuffled	0	76	80	47.42	631
GPS shuffled	1	46	56	22.15	630
GPS shuffled	2	6	16	6.00	634
GCN	0	8	18	1.26	524
GCN	1	8	18	0.98	522
GCN	2	9	19	1.36	522
Local MLP	0	8	18	2.20	497
Local MLP	1	77	80	5.30	497
Local MLP	2	74	80	6.21	497
Mean	0	8	18	1.27	521
Mean	1	8	18	1.01	522
Mean	2	9	19	1.24	522
GraphSAGE	0	48	58	6.51	570
GraphSAGE	1	60	70	3.77	570
GraphSAGE	2	39	49	2.81	570

Diagnostic interpretation. Undefined correlations remain undefined, with valid-gene counts in the exports. Raw Spearman is secondary because near-tied ranks are floating-point sensitive. Distance curves retain occupied-bin support; different bins need not contain the same locations or patients. Conditional geometric checks and remote-feature sensitivity do not establish biological association. Degree/length controls with zero accepted swaps do not

provide an independent rewired comparison. Optional anchor-conditioned learning and full-image equivariance tests were not performed.

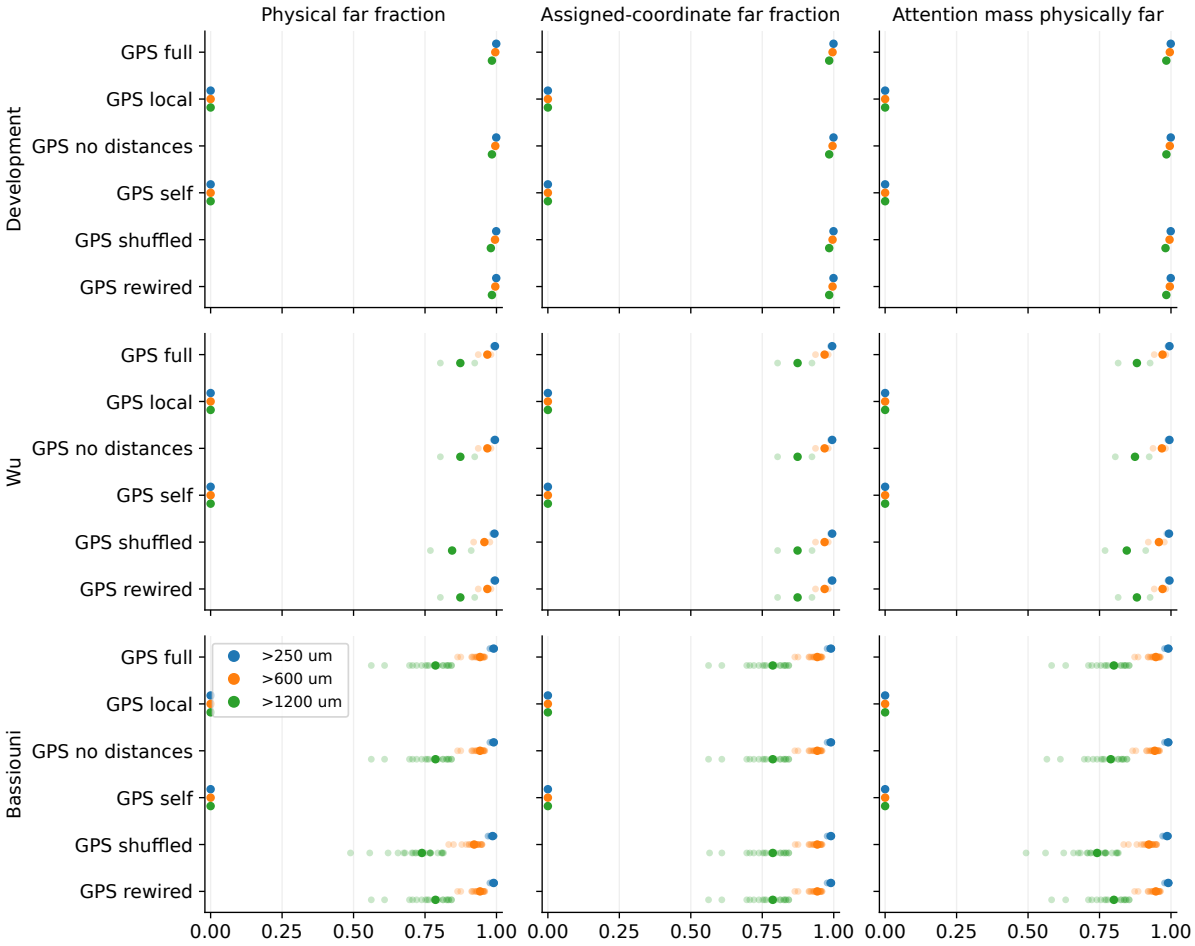


Figure S24: Conditional attention reach. Dots are individual patient summaries after equal seed/head/layer means; large dots are cohort medians. Colors identify 250/600/1200 micrometer thresholds. This is a software/connectivity diagnostic, not evidence of molecular correlation. All GPS variants, seeds and patients are exported.

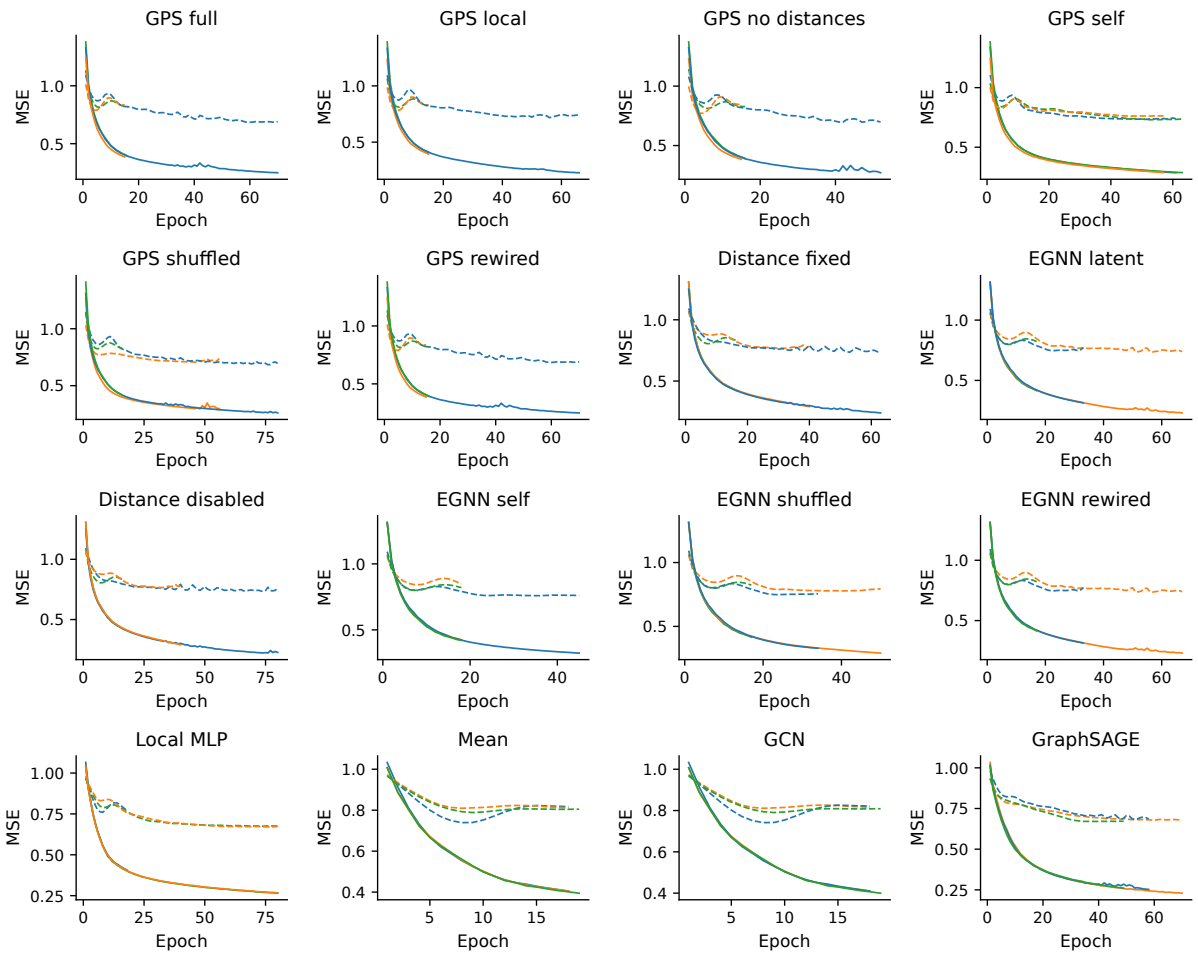


Figure S25: All 48 source-only training histories. Every one of 16 configurations and all three seeds are retained. Solid curves show training MSE, dashed curves validation MSE. Blue, orange and green identify seeds 0, 1 and 2, respectively. Only TENX95 validation MSE selected epochs; evaluation outcomes did not select architecture or seeds.

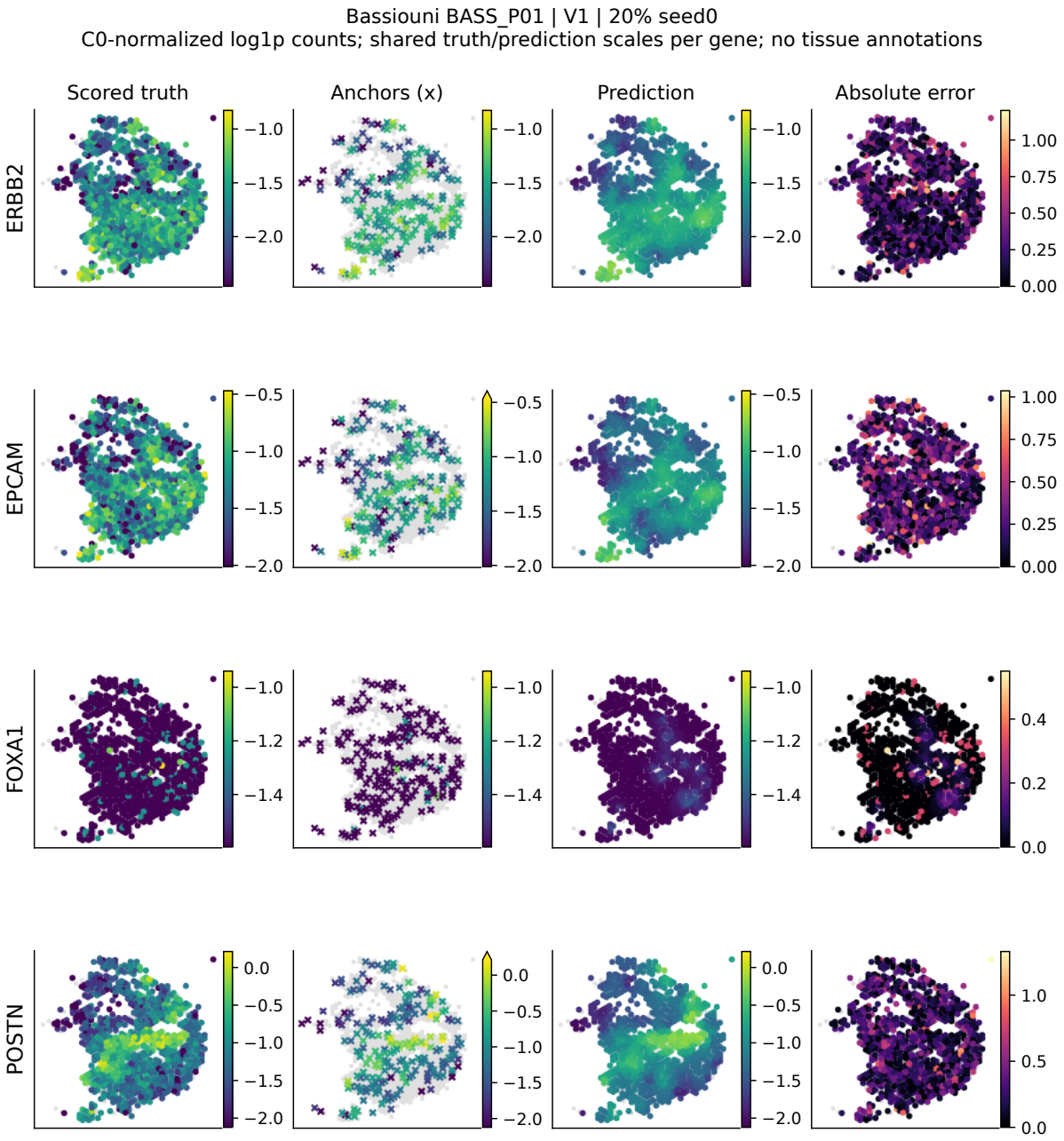


Figure S26: Predetermined Bassiouni distributed map. BASS P01 at 20percent seed0, the four prespecified genes, original IDW k16/p2. Truth and prediction share each gene scale; anchor colorbar extensions mark clipped anchor values. Gray sites are unscored context. Constant fields are retained and explicitly marked; no tissue annotations are inferred.

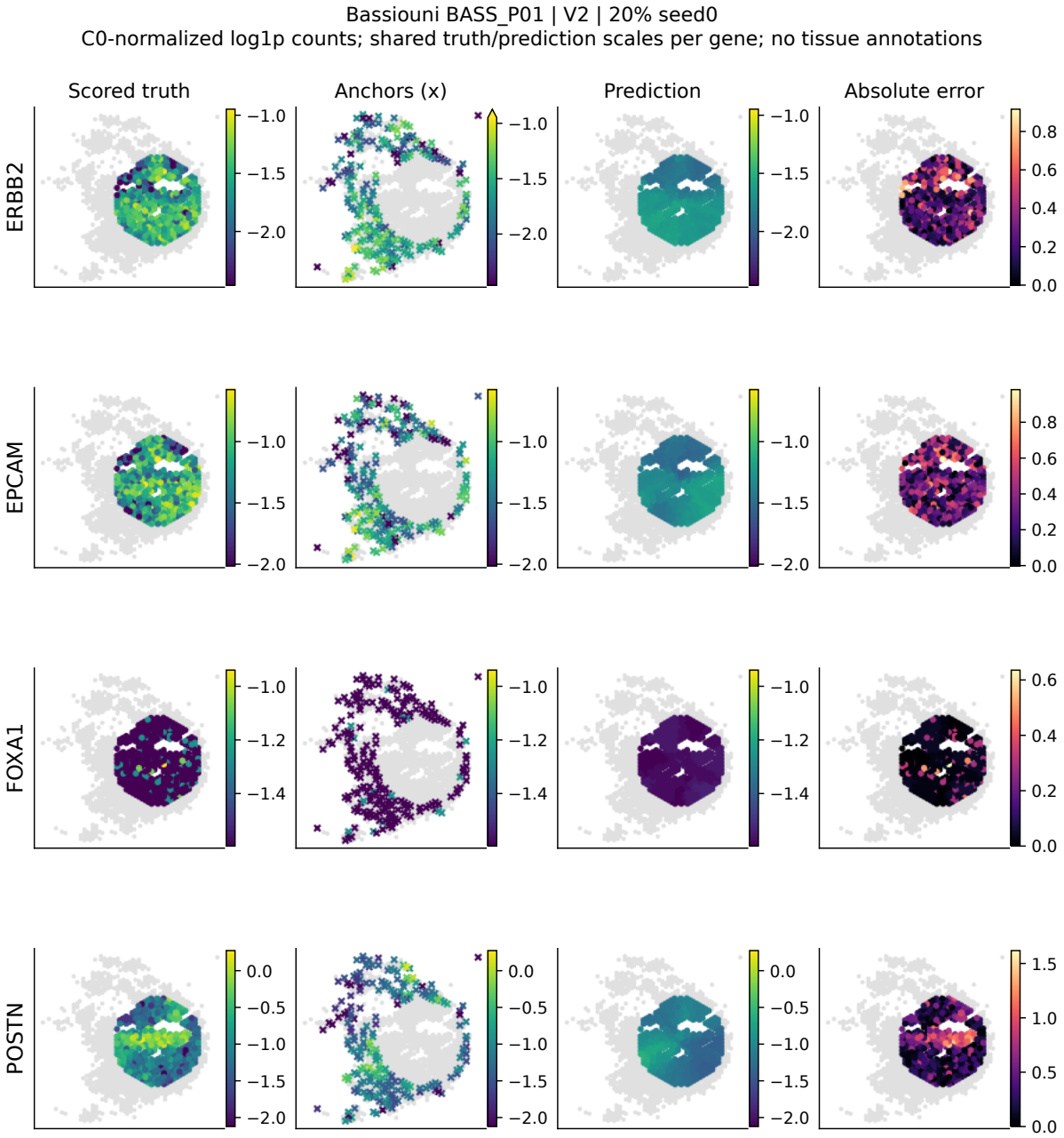


Figure S27: Predetermined Bassiouni regional map. BASS P01 central V2 at 20percent seed0, uniform k16 and the same four predetermined genes. Historical V2 has different scored support and estimator from V1; this map comparison does not isolate geometry. Identical-H effects are evaluated separately. Truth/prediction scales are shared within each gene, errors use a separate nonnegative scale.

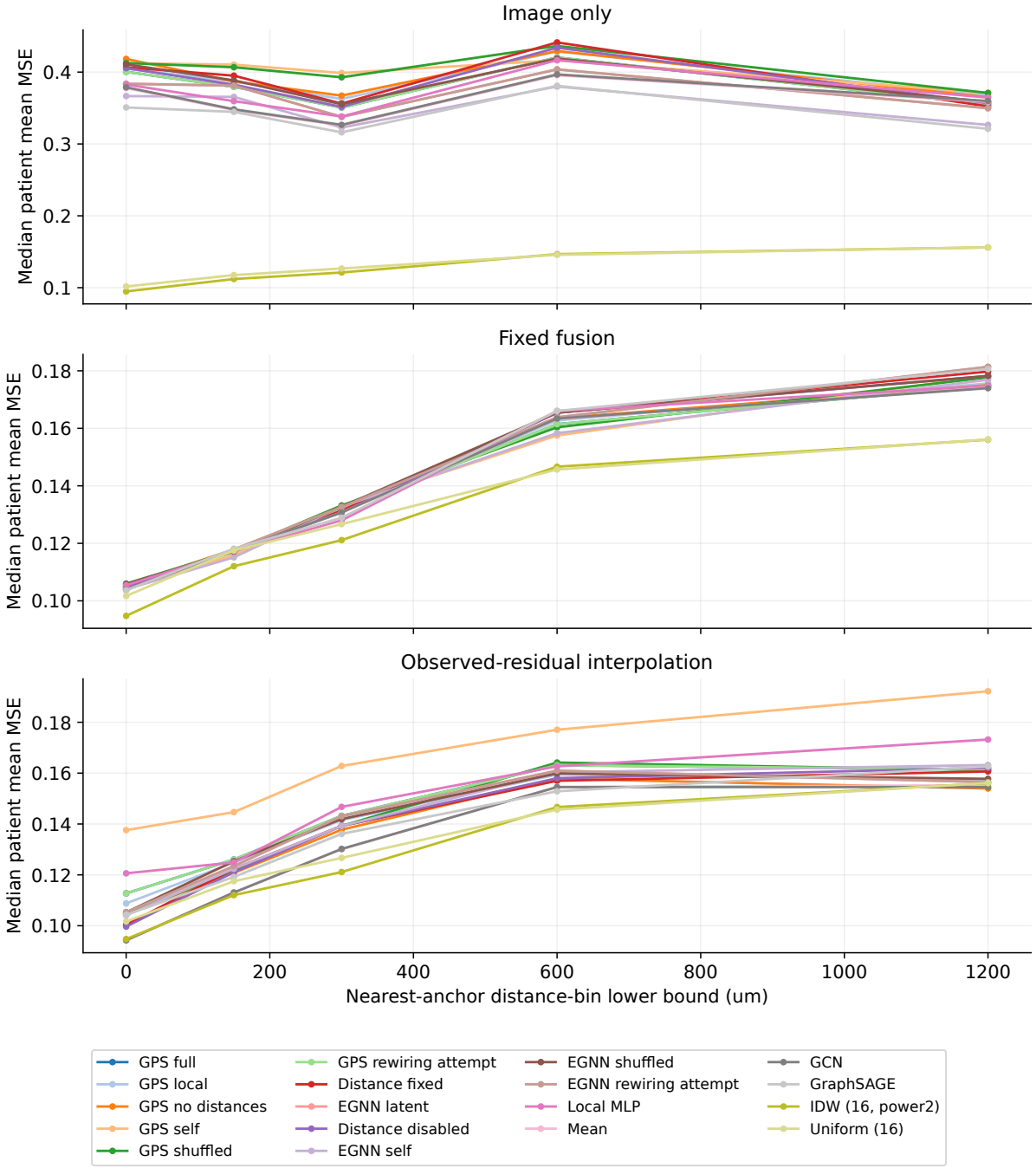


Figure S28: Distance-resolved Bassiouni reconstruction. All16 new models and both molecular baselines,20percent region-exclusion B2. Curves are medians of per-patient conditional-bin mean errors after equal mask and training-seed averages. Occupied bins only; empty bins are omitted. Source tables retain number of patients and min/median/max mean locations per mask in each bin. Different bins need not contain the same patients or locations and do not identify a biological correlation length.

Table S17: All 48 full-section development-test fits, with every training seed retained. TENX99 at 20% coverage, averaging the same three original masks within each seed. Left metrics are image-only; right metrics use exploratory residual anchors. This is one previously inspected patient, not 48 biological replicates. No confidence interval or seed selection is inferred. Full precision, fixed-fusion, all budgets, external patient metrics and older frozen heads accompany the CSV exports.

Model	Seed	Image r	MSE	MAE	Residual r	MSE	MAE
GPS full	0	0.714	0.9751	0.7814	0.834	0.2349	0.3694
GPS full	1	0.529	1.1646	0.8594	0.801	0.2693	0.3967
GPS full	2	0.569	1.2240	0.8969	0.813	0.2634	0.3909
GPS local	0	0.708	0.9202	0.7560	0.826	0.2662	0.3921
GPS local	1	0.573	1.1372	0.8470	0.804	0.2694	0.3958
GPS local	2	0.508	1.2035	0.8928	0.811	0.2617	0.3885
GPS no distances	0	0.711	1.0426	0.8070	0.835	0.2242	0.3589
GPS no distances	1	0.556	1.1672	0.8560	0.809	0.2596	0.3878
GPS no distances	2	0.577	1.2357	0.8978	0.819	0.2545	0.3827
GPS self	0	0.664	1.0495	0.8149	0.803	0.3344	0.4438
GPS self	1	0.669	1.0613	0.8177	0.795	0.3441	0.4500
GPS self	2	0.671	1.0259	0.8074	0.808	0.3001	0.4204
GPS shuffled	0	0.717	0.9751	0.7806	0.822	0.2865	0.4111
GPS shuffled	1	0.710	1.2113	0.8818	0.829	0.2456	0.3793
GPS shuffled	2	0.552	1.2417	0.9061	0.812	0.2733	0.4001
GPS rewired	0	0.714	0.9751	0.7814	0.834	0.2349	0.3694
GPS rewired	1	0.529	1.1646	0.8594	0.801	0.2693	0.3967
GPS rewired	2	0.569	1.2240	0.8969	0.813	0.2634	0.3909
Distance fixed	0	0.680	0.9648	0.7772	0.820	0.2945	0.4164
Distance fixed	1	0.637	0.9801	0.7883	0.793	0.3136	0.4327
Distance fixed	2	0.471	1.1048	0.8520	0.793	0.2867	0.4118
EGNN latent	0	0.628	1.0248	0.8072	0.799	0.3033	0.4259
EGNN latent	1	0.711	0.9576	0.7706	0.825	0.2646	0.3915
EGNN latent	2	0.536	1.0697	0.8344	0.792	0.3030	0.4239
Distance disabled	0	0.706	1.0438	0.8028	0.831	0.2609	0.3887
Distance disabled	1	0.640	0.9790	0.7877	0.794	0.3128	0.4320
Distance disabled	2	0.471	1.1047	0.8519	0.793	0.2866	0.4118
EGNN self	0	0.595	0.9884	0.7966	0.778	0.3307	0.4476
EGNN self	1	0.546	1.1053	0.8471	0.791	0.3008	0.4227
EGNN self	2	0.545	1.0548	0.8283	0.793	0.3046	0.4253
EGNN shuffled	0	0.627	1.0708	0.8276	0.794	0.3082	0.4303
EGNN shuffled	1	0.665	1.0025	0.7999	0.785	0.3441	0.4550
EGNN shuffled	2	0.532	1.0732	0.8359	0.791	0.3038	0.4245
EGNN rewired	0	0.628	1.0248	0.8072	0.799	0.3033	0.4259

Model	Seed	Image r	MSE	MAE	Residual r	MSE	MAE
EGNN rewired	1	0.711	0.9576	0.7706	0.825	0.2646	0.3915
EGNN rewired	2	0.536	1.0697	0.8344	0.792	0.3030	0.4239
Local MLP	0	0.604	0.9930	0.8144	0.829	0.2433	0.3775
Local MLP	1	0.669	0.9033	0.7628	0.820	0.2591	0.3910
Local MLP	2	0.667	0.9185	0.7714	0.811	0.2571	0.3897
Mean	0	0.624	1.0157	0.8231	0.827	0.2361	0.3667
Mean	1	0.593	1.0522	0.8471	0.819	0.2387	0.3688
Mean	2	0.627	1.0723	0.8525	0.824	0.2366	0.3674
GCN	0	0.620	1.0175	0.8241	0.825	0.2386	0.3692
GCN	1	0.591	1.0536	0.8478	0.818	0.2406	0.3707
GCN	2	0.623	1.0727	0.8529	0.821	0.2385	0.3694
GraphSAGE	0	0.671	0.8701	0.7488	0.827	0.2333	0.3675
GraphSAGE	1	0.688	0.8818	0.7530	0.829	0.2337	0.3683
GraphSAGE	2	0.667	0.9245	0.7785	0.821	0.2307	0.3653

S6 Detailed development methods

These specifications distinguish the original development comparators from the additional minibatch and full-section comparisons. They retain the model-selection and calibration details needed to reproduce the corresponding figures.

S6.1 Development head searches and conditional patient holdouts

The original development MLPs used one hidden linear layer, GELU, dropout and a linear gene-output layer. Features were standardized on fitting patients before graph aggregation; a separate fitting-only head-target scaler was inverted before scoring. Population scales below 10^{-8} were replaced by 1. AdamW used learning rate 10^{-3}, weight decay 10^{-4}, minibatches of 512 with shuffled rows and retained final partial batches, at most 80 epochs, patience 10 and minimum validation-MSE improvement 10^{-4}. Best qualifying parameters were restored, with dropout disabled at prediction. Three training seeds 0/1/2 were retained; hyperparameter candidates used seed 0. Computation used deterministic CPU float32 tensors and four threads, without a learning-rate schedule or auxiliary loss. These settings describe the original development comparisons below, not the later full-section regime.

The initial local comparison in Supplementary Fig. S3 selected hidden width from $\{128, 256\}$ and dropout from $\{0, 0.1\}$ by the lowest validation MSE. For 50 genes, the selected architecture was $512 \to 128 \to 50$ with dropout 0.1. The ridge baseline used an intercept and a Cholesky solver, with penalty selected from $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$ by the highest validation median gene Pearson correlation, breaking ties by lower MSE and then smaller penalty; the selected 50-gene penalty was 100. Training-target and training-embedding permutations disrupted pairing within the fitting data; validation and evaluation data were unchanged.

The representation comparison in Supplementary Fig. S6 used the same four width/dropout candidates but selected the highest validation median gene Pearson correlation, with lower MSE as the first tie-breaker. Both ResNet18 and Phikon selected width 256 and dropout 0.1 for 50 genes. Thus the initial local predictor in Supplementary Fig. S3 and the ResNet18 predictor used in the representation comparison and original sparse pipeline have different selected widths. The latter predictor is described in the main Methods. The representation comparison also evaluated ridge penalties $\{0.1, 1, 10, 100, 1000\}$; both encoders selected 1000 for 50 genes. Equal width and search budget did not imply equal parameter count because the input dimensions differed.

Conditional leave-one-patient-out sensitivity fits used the remaining three development patients, refitted feature and head-target scalers on those patients, retained the primary-selected hyperparameters and used fixed epoch counts without early stopping on the held-out patient. For the 50-gene heads, these counts were 14 for the initial MLP, 14 for mean pooling and GCN, 7 for GraphSAGE, 12 for the representation-comparison ResNet18 head and 1 for the Phikon head. These were rounded medians of the corresponding primary fits' best epochs. Since the configurations and underlying target definition had already been selected during development, these fits are conditional sensitivity analyses.

S6.2 Original graph controls and pathology representation

The matched controls in Supplementary Fig. S5 retained the GraphSAGE head and training settings. The self-only control used $[X, X]$. Vertex permutation preserved the graph's degree distribution, while double-edge rewiring preserved each node's degree. The additional distance-constrained rewiring preserved the numbers of physical edge lengths in $[0, 175)$, $[175, 250)$ and $[250, \infty)$ μm. Rewiring targeted five accepted swaps per edge, with limits of 100 attempts per edge for degree-only rewiring and 1,000 for distance-constrained rewiring; the latter used candidate endpoints within 275 μm. Five graph randomization seeds were crossed with the three model seeds. The Phikon spatial sensitivity comparison reused the selected graph, hidden widths and dropout: mean pooling used $768 \to 128 \to 50$, and GraphSAGE used $1536 \to 67 \to 50$, with true and degree-plus-length-matched graphs. These inherited widths do not constitute exact parameter matching between encoders.

The historical Phikon representation is defined in the main Methods subsection "Registration and frozen image features". Its C4 extraction used float32 evaluation-mode inference, batches of 16 and eight CPU threads (Transformers 4.57.3 and PyTorch scaled dot-product attention). It used the same archived RGB patches without recropping. Historical head selection and target scaling followed the development procedures above; these differ from the direct-C0-target, validation-MSE contract of the matched-backbone follow-up. Neither analysis fine-tuned the encoder.

S6.3 Development ridge residual correction

The learned-propagation model in Supplementary Fig. S7 is a gene-specific ridge residual correction using fixed graph summaries. Its image prediction H is the equal average of the three frozen local MLP predictions from the representation comparison. Let o be the observed-location indicator and Z the expression matrix containing permitted observed vectors and zeros elsewhere. A self-inclusive row-normalized operator P was constructed from the symmetrized six-nearest-neighbor graph capped at 250 μm. The observed masses and molecular messages were

$$m^{(1)} = Po, \quad m^{(2)} = P^2 o, \qquad V_{ig}^{(1)} = \frac{(PZ)_{ig}}{\max(m_i^{(1)}, 10^{-6})}, \quad V_{ig}^{(2)} = \frac{(P^2 Z)_{ig}}{\max(m_i^{(2)}, 10^{-6})}.$$

Image features were standardized on fitting patients and reduced to eight whitened principal components with randomized singular-value decomposition and random seed 0. Denote these components by X_{PCA}. The 29 shared covariates at each location were X_{PCA}, PX_{PCA}, $P^2 X_{\mathrm{PCA}}$, o, $m^{(1)}$, $m^{(2)}$, $\log(1 + d_1/100)$, and the fraction of the nearest 16 observed anchors lying within 300 μm, where d_1 is nearest-anchor distance in μm. Five further covariates for gene g were the image prediction H_g, the fixed fusion prediction F_g, Z_g, $V_g^{(1)}$ and $V_g^{(2)}$. The resulting design has 34 covariates and an intercept for each gene.

Predictions were $\widehat{y}_g = F_g + \Phi_g \theta_g$, where the last column of Φ_g is one. Coefficients were fitted by

$$\underset{\theta_g}{\operatorname{argmin}} \ \|(y_g - F_g) - \Phi_g \theta_g\|_2^2 + \alpha \, \|\theta_{g,1:34}\|_2^2 .$$

The intercept was unpenalized. Fitting sampled up to 256 hidden locations without re-

placement per training mask using that mask's seed; their expression entered only the supervised response. Masks covered 5, 10, 20, 40, 60 and 80% of locations under random, block and capture-window sampling, with three replicates per condition. The penalty grid was $\{1, 100, 10000\}$, selected by mean validation MSE across masks with equal mask weights; both encoders selected $\alpha = 10000$ for 50 genes. No additional expression scaler or auxiliary loss was introduced.

The underlying development fusion was $F = \lambda H + (1 - \lambda)M$. Nearest-anchor mean interpolation selected k from $\{1, 4, 8, 16\}$; inverse-distance interpolation used up to 16 anchors and selected power from $\{1, 2, 3\}$. The stronger family and $\lambda \in \{0, 0.25, 0.5, 0.75, 1\}$ were chosen by mean validation MSE separately by panel and sampling geometry. For 50 genes, random sampling selected inverse-distance-squared interpolation and $\lambda = 0.25$; block sampling selected the 16-anchor mean and $\lambda = 0.5$; capture-window sampling selected inverse-distance-cubed interpolation and $\lambda = 0.5$. These development settings differ from the final external fusion settings. Supplementary Fig. S7 evaluates three permutation seeds on one random 20% observation mask, with the fitted models unchanged. The plotted B1, B3 and B4 labels denote the frozen image predictor, fixed fusion and ridge residual correction, respectively.

S6.4 Patient-disjoint HER2ST calibration and fusion selection

Supplementary Figs. S1 and S10 use the same eight HER2ST patients and 40 shared genes. Four outer folds held out the pairs A1/B1, C1/D1, E1/F1 and G1/H1, respectively. Within each remaining six-patient set, target-mapping candidates were evaluated by leave-one-patient-out records: the mapping for an inner patient's predictions was estimated from the other five patients. The fixed source moments came from NCBI783 and NCBI785. The candidates were the unchanged target, per-gene mean/standard-deviation mapping and per-gene median/interquartile-range mapping. Each patient had equal total weight in moment or quantile estimates. An interquartile range below 10^{-8} fell back to the corresponding standard deviation, and a standard deviation below 10^{-8} was replaced by one. Source and destination moments were estimated separately. Image predictions were expressed in source-standardized coordinates, anchors in destination-standardized coordinates, and final errors were computed after mapping predictions back to common target units.

The target-mapping comparison retained fixed image weights 0.25 for V1 and 0.5 for V2. Selection used inner random-mask seeds 0–2 and four equally weighted conditions: V1 coverage 10, 20 and 40%, and V2 coverage 20%. Seeds and geometries were first averaged within patient and condition, then conditions within patient, then the six patients. Candidates within 0.005 of the highest mean patient median gene Pearson correlation were eligible; among these, the candidate with lowest common-unit MSE was selected. The mean/standard-deviation transform was selected in every fold; the corresponding outer mapping used moments estimated from all six outer-training patients.

The selected-fusion analysis then used the corresponding inner records to fit and select among four families: molecular interpolation alone; one global image weight from $\{0, 0.1, \ldots, 1\}$; 35 nondecreasing three-bin weight rules with weights from $\{0, 0.1, 0.25, 0.5, 1\}$ and nearest-anchor boundaries 300 and 600 μm; and a five-parameter sigmoid gate. All weights were shared

across genes, regimes and budgets within an outer fold. The sigmoid gate was

$$w_i = \mathrm{sigmoid}(\beta_0 + \widetilde{q}_i^{\top}\beta), \qquad \widehat{y}_i = M_i + w_i(H_i - M_i),$$

where the four features before fitting-data standardization were $\log(1+d_1/200)$, $\log(1+\overline{d}_{16}/200)$, $\log(1+n_{600})$ and $\log(1+v)$: nearest-anchor distance, mean distance to up to 16 nearest anchors, the number of observed anchors within 600 μm, and mean across-gene variance among the neighboring observed vectors. The gate minimized common-unit squared error with equal patient/condition/run/location contributions, divided by each run's interpolation MSE floored at 10^{-8}, plus $0.01\|\beta\|_2^2 + 10^{-6}\beta_0^2$. The same weights defined feature means and standard deviations. L-BFGS-B used initial coefficients $(-2, 0, 0, 0, 0)$, bounds $[-12, 12]$, maximum 300 iterations, function tolerance 10^{-12} and gradient tolerance 10^{-7}. No outer-patient outcomes entered fitting or candidate selection.

The same correlation-then-MSE rule selected the fusion family. The first three patient-pair folds selected global image weight 0.2; the G1/H1 fold selected the sigmoid gate. Outer evaluation used mask seeds 0–9. Supplementary Fig. S10 reports these fold-selected rules, whereas Supplementary Fig. S1 shows the fixed-fusion target-mapping comparison. Both remain calibration-development analyses of previously examined HER2ST patients.

The full-precision primary image feature scaler, target normalization and source-to-Wu calibration constants are supplied in `methods_data/`. The primary head settings and recorded software versions are provided there in machine-readable form. These constants specify the published transformations without refitting them.

S6.5 Identity controls and numerical audit scope

Local, mean-self and GCN-self inputs in the 30-head comparison were bitwise identical. Seeds 1/2 predictions also agreed bitwise; seed 0 weights differed by at most 2.98×10^{-8} and predictions by 9.54×10^{-7}, within the stated float32 guard. These are computational identity checks, not independent biological support. At 20% coverage, matched-seed GCN spatial-minus-self correlation changes ranged from -0.0425 to -0.0149; GraphSAGE changes ranged from $+0.0002$ to $+0.0318$. Shuffled GraphSAGE seed-mean image prediction exceeded the spatial version in TENX99. These development findings do not establish a general topology advantage.

The original Bassiouni evaluation produced 2,904 clean and 18 control bundles (8,766 method predictions); all 17,532 hidden-label substitutions were invariant. The recorded audit reproduced 8,766 predictions and 35,064 per-gene metric arrays and reconstructed 968 label-free mask trajectories. Additional Wu molecular comparisons comprised 2,700 runs and 5,400 exact substitutions; all 360 original random references and 54,000 primary gene values passed 10^{-10} tolerances. Additional Bassiouni comparisons comprised 19,800 runs and 39,600 exact substitutions, with 2,640 original random references and 396,000 primary per-gene values passing the same tolerances. The neural matrix retained 2,028 model–patient bundles and 289,224 exact hidden-label checks; all 384 conditional geometric checks passed. Raw Spearman is retained but excluded from exact-replay claims because of near ties. Hashes, protocols, complete results and environment records provide the detailed provenance.

S7 Matched-backbone records and reproducibility

The matched comparison distinguishes graph context relative to a local visual head from improvement beyond the same molecular anchors. Main Methods provide the authoritative definitions of image preprocessing, graph operators, training, calibration, masks, reconstruction and patient pairing. The machine-readable comparison report separates these two questions by backbone, route and cohort, retaining the single development patient and all three individual Wu effects. The original 22-patient placement and 20-estimator studies remain separate sources of evidence.

The initial fixed-rate ResNet18 analysis reuses verified states, predictions and per-run/per-gene exports. Its results are preserved as a distinct sensitivity analysis. The two-rate refinement followed source/validation learning-curve inspection before any Phikon evaluation outcomes; it gave each graph and non-graph configuration identical search and selection opportunities. All candidate losses, seeds, histories and state hashes remain in the candidate manifest, while the selected manifest was committed before selected-model scoring. These source/validation diagnostics are not external performance evidence.

The primary condition uses three historical masks at 20% distributed coverage, not the ten masks underlying the original molecular headline correlations. Every paired predictor uses identical observation and scoring supports, recorded with barcode, coordinate, gene, target and index hashes. Supporting files retain 10/40% coverage, identical-target placement and the historical central regional mask. Historical regional completion and identical-target placement remain distinct designs.

Per-run/per-gene exports retain undefined Pearson values and raw Spearman scores. The common-valid-gene sensitivity is explicitly separate from the primary defined-gene metric. Mask and training seeds represent computational variation; biological pairing and Bassiouni resampling use patients. New selected-model scoring repeats NaN/random corruption of all unobserved labels, including unscored locations, with exact unchanged hybrid predictions, and verifies image-only anchor independence on fixed hidden support. Reused checks are identified as verified prior evidence rather than newly executed tests.

Registered RGB patch bytes are identical across matched heads. Across encoders, pretraining, embedding dimension and effective physical field differ, as described in the main Methods. No interpolation-based enlargement restores native image detail. Frozen feature extraction is distinct from head fitting and from encoder fine-tuning. New derived states and complete numerical bundles remain author-local; manifests do not constitute public deposition. The three original primary image-head binaries remain missing.

Table S18: Matched-backbone fitting and validation selection. Each configuration used the same two candidate learning rates (0.001 and 0.0001) with seeds 0/1/2. One rate was selected per backbone/configuration using mean best qualifying TENX95 MSE across all three seeds; no evaluation outcome selected it. Epoch triplets retain seed order 0/1/2. Updates count all minibatches, including the final partial batch. Times sum the three selected fits; peak RSS is their maximum and includes loaded data. These observed execution costs are not a controlled speed comparison across encoders or historical execution sessions. All 72 candidate histories, validation losses, state/configuration/input hashes and selected flags are machine-readable.

Backbone	Head/control	Parameters	Rate	Best epochs	Run epochs	Updates	Seconds / MiB
ResNet18	local	72,114	0.0010	12/14/17	22/24/27	1022	11.5 / 650
ResNet18	mean	72,114	0.0010	12/16/14	22/26/24	1008	9.9 / 584
ResNet18	GCN	72,114	0.0010	12/14/14	22/24/24	980	8.6 / 637
ResNet18	GraphSAGE	72,075	0.0010	7/10/7	17/20/17	756	7.9 / 657
ResNet18	GCN shuffled	72,114	0.0010	1/9/15	11/19/25	770	7.0 / 644
ResNet18	GraphSAGE shuffled	72,075	0.0010	9/18/7	19/28/17	896	10.8 / 687
Phikon	local	104,882	0.0001	5/7/6	15/17/16	672	7.5 / 681
Phikon	mean	104,882	0.0001	5/7/6	15/17/16	672	9.1 / 681
Phikon	GCN	104,882	0.0001	5/7/6	15/17/16	672	9.3 / 681
Phikon	GraphSAGE	104,792	0.0010	1/1/1	11/11/11	462	4.5 / 730
Phikon	GCN shuffled	104,882	0.0001	7/7/7	17/17/17	714	11.6 / 704
Phikon	GraphSAGE shuffled	104,792	0.0001	11/12/12	21/22/22	910	12.6 / 768